200C - Project Estimation

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```
#select treatment

#estimation_data$treat <- estimation_data$etv_dh
estimation_data$treat <- estimation_data$etv_all_jep
#estimation_data$treat <- estimation_data$etv_cnmh
# estimation_data$treat <- estimation_data$etv_all_cnmh
# estimation_data$treat <- estimation_data$etv_false_pos

#dichotomized treatment
estimation_data <- estimation_data |>
  mutate(treat_dicho = ifelse(treat > 0, 1, 0))

#missings
#TODO: NA's found in data input.  IT IS HIGHLY RECOMMENDED THAT YOU TEST IF THE MISSING VALUES ARE BALA

estimation_data |>
  summarise_all(~ sum(is.na(.))) #how many missing for each
```

```
## # A tibble: 1 x 30
##    dpto  mpio codmpio trust_plebi_yes etv_dh etv_all_jep etv_false_pos etv_cnmh
##   <int> <int>   <int>           <int>  <int>       <int>         <int>    <int>
## 1     0     0       0               2      0           0             0        0
## # i 22 more variables: etv_all_cnmh <int>, uribe_2002_proportion <int>,
## #   uribe_2006_proportion <int>, avg_schooling <int>, literacy <int>,
## #   pop_total <int>, prop_rural <int>, gdp_pcap_cons2005 <int>, mpi <int>,
## #   gini <int>, log_pop <int>, etv_state_pre <int>, etv_nonstate_pre <int>,
## #   presence_nonstate_pre <int>, etv_guerr_pre <int>, presence_guerr_pre <int>,
## #   etv_param_pre <int>, presence_param_pre <int>, dem_security <int>,
## #   hist_land_conflict <int>, treat <int>, treat_dicho <int>
```

```
estimation_nona <- estimation_data |>
  drop_na() #1115 obs to 1006 - which ones?

#covariates
covariates <- c("uribe_2002_proportion", "uribe_2006_proportion", "avg_schooling", "literacy", "log_pop

balance_formula <- as.formula(paste("treat_dicho ~ ", paste(covariates, collapse=" + ")))

# calculate propensity scores
estimation_nona$ps <- stats::glm(balance_formula,
                                 data = estimation_nona,
                                 family = binomial(link = logit)
```

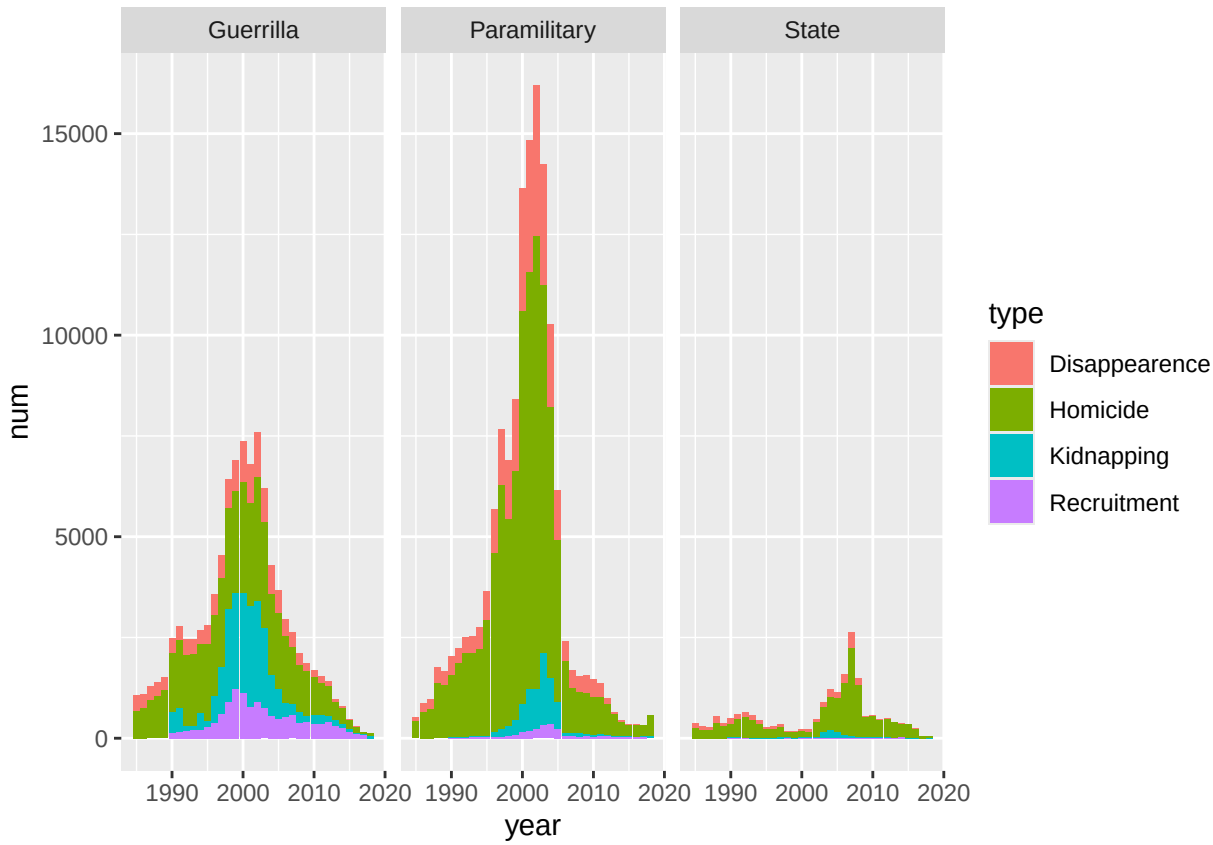
```
)$fitted.values
```

1. Descriptive statistics

1.1 Histogram for ETV (time and number of incidents, color by actor faceted by type)

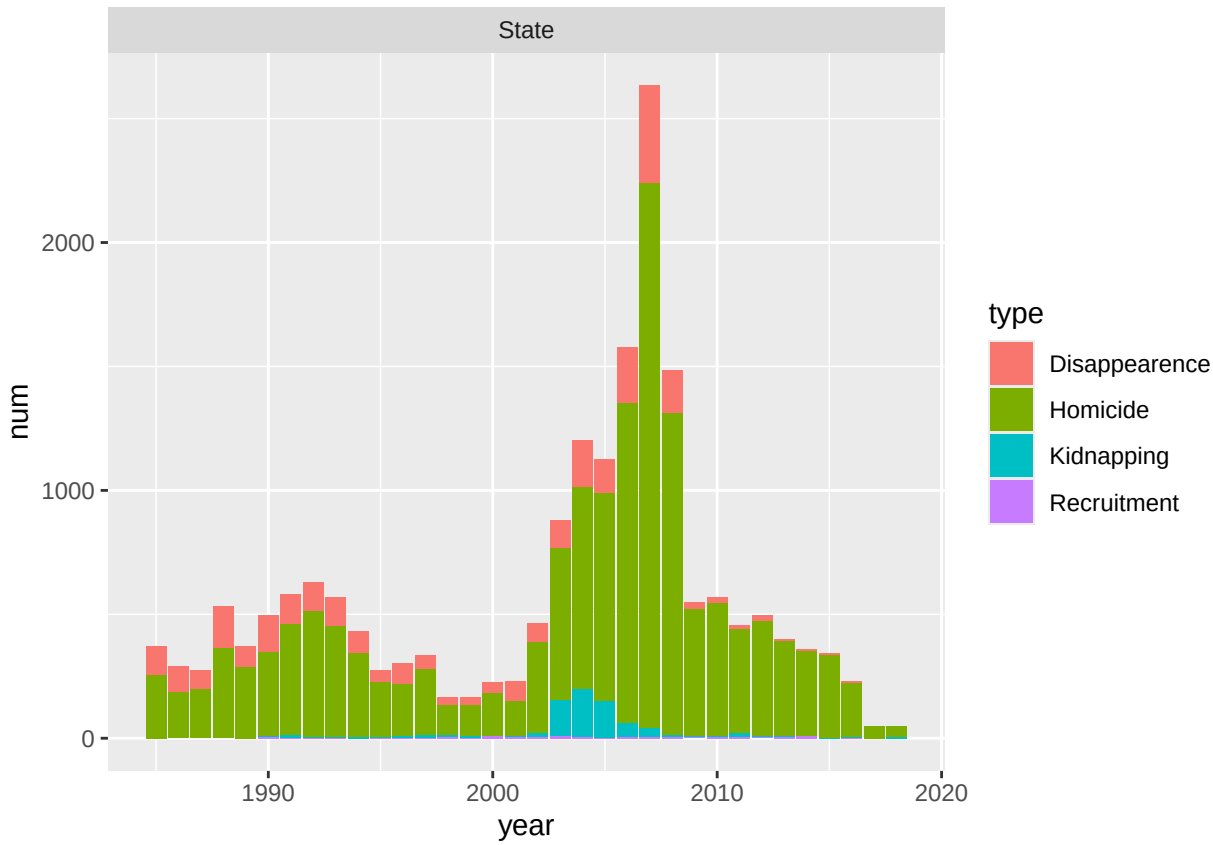
```
full_dataset |>
  dplyr::select(contains(c("year", "_jep"))) |> #types
  group_by(year) |> #time
  pivot_longer(-year, names_to = "jep", values_to = "num") |> #get column of count
  mutate(
    perp = case_when( #make perpetrator column
      grepl("Guerr", jep) ~ "Guerrilla",
      grepl("Param", jep) ~ "Paramilitary",
      grepl("State", jep) ~ "State",
      grepl("Other", jep) ~ "Other",
      grepl("total", jep) ~ "Total",
      grepl("Multiple", jep) ~ "Multiple"
    )
  ) |>
  mutate(
    type = case_when( #make violence type column
      grepl("Hom", jep) ~ "Homicide",
      grepl("Dis", jep) ~ "Disappearence",
      grepl("Kid", jep) ~ "Kidnapping",
      grepl("Rec", jep) ~ "Recruitment"
    )
  ) |>
  filter(!perp %in% c("Total", "Other", "Multiple")) |> #remove the total
  ggplot(aes(x = year, y = num, fill = type)) + #plot count by year; type by color
    geom_col() +
    facet_wrap(~perp)
```

```
## Warning: Removed 368668 rows containing missing values or values outside the scale range
## (`geom_col()`).
```




```
full_dataset |>
  dplyr::select(contains(c("year", "_jep"))) |> #types
  group_by(year) |> #time
  pivot_longer(-year, names_to = "jep", values_to = "num") |> #get column of count
  mutate(
    perp = case_when( #make perpetrator column
      grepl("State", jep) ~ "State"
    )
  ) |>
  filter(perp == "State") |>
  mutate(
    type = case_when( #make violence type column
      grepl("Hom", jep) ~ "Homicide",
      grepl("Dis", jep) ~ "Disappearence",
      grepl("Kid", jep) ~ "Kidnapping",
      grepl("Rec", jep) ~ "Recruitment"
    )
  ) |>
  ggplot(aes(x = year, y = num, fill = type)) + #plot count by year; type by color
    geom_col() +
    facet_wrap(~perp)
```

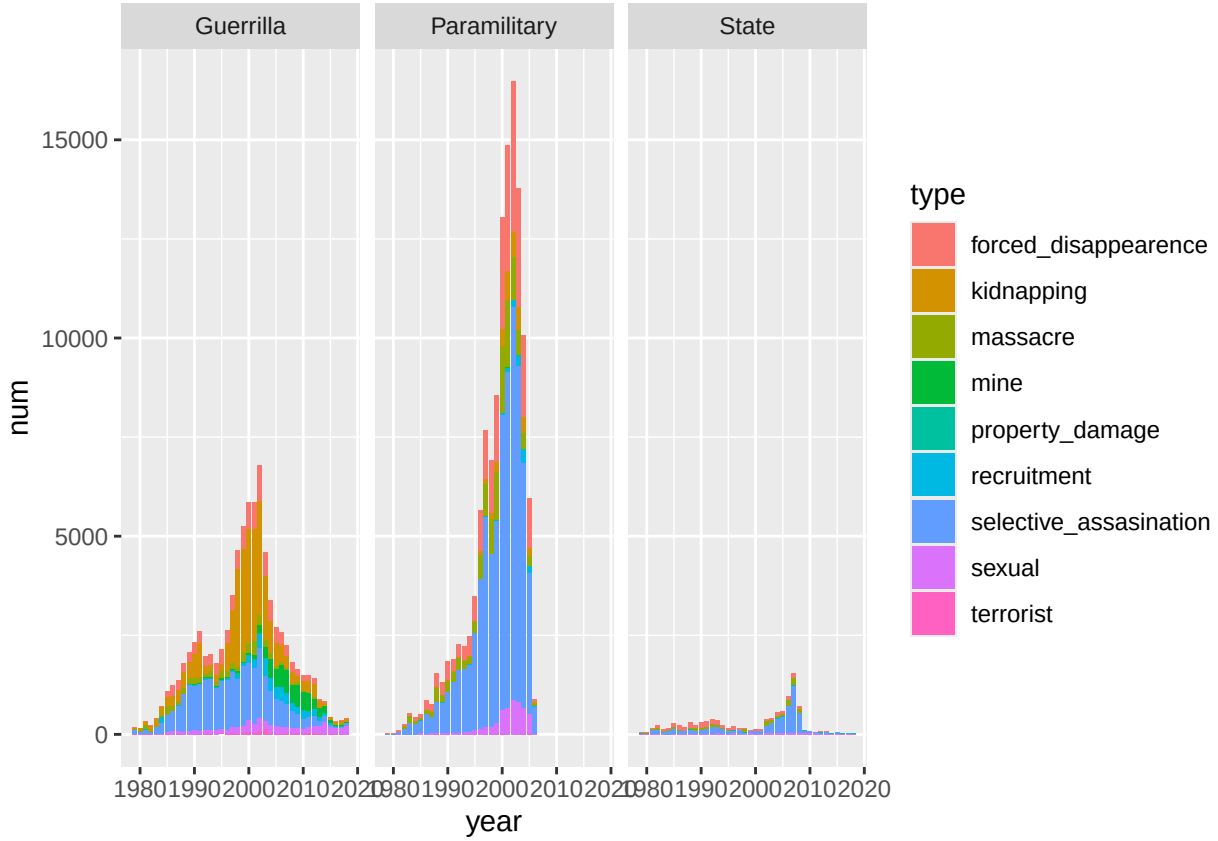
```
## Warning: Removed 134176 rows containing missing values or values outside the scale range
## (`geom_col()`).
```




```
full_dataset |>
  dplyr::select(contains(c("year", "_cnmh"))) |>
  group_by(year) |> #time
  pivot_longer(-year, names_to = "cnmh", values_to = "num") |> #get column of count
  mutate(
    perp = case_when( #make perpetrator column
      grepl("guerrilla", cnmh) ~ "Guerrilla",
      grepl("paramilitary", cnmh) ~ "Paramilitary",
      grepl("state", cnmh) ~ "State"
    )
  ) |>
  mutate(
    type = case_when( #make violence type column
      grepl("forced_disappearence", cnmh) ~ "forced_disappearence",
      grepl("kidnapping", cnmh) ~ "kidnapping",
      grepl("massacre", cnmh) ~ "massacre",
      grepl("property_damage", cnmh) ~ "property_damage",
      grepl("mine", cnmh) ~ "mine",
      grepl("recruitment", cnmh) ~ "recruitment",
      grepl("selective_assasination", cnmh) ~ "selective_assasination",
      grepl("sexual", cnmh) ~ "sexual",
      grepl("terrorist", cnmh) ~ "terrorist"
    )
  ) |>
  ggplot(aes(x = year, y = num, fill = type)) + #plot count by year; type by color
    geom_col() +
```

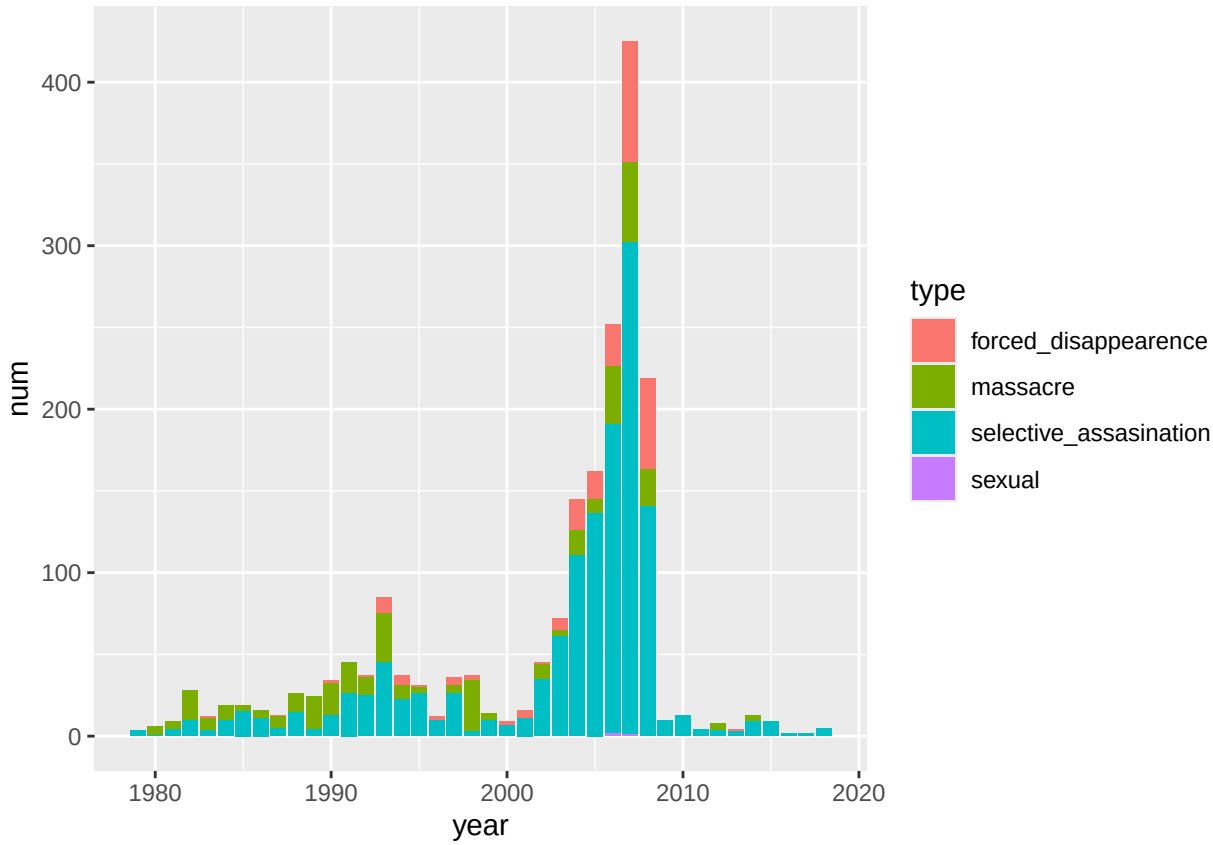
```
    facet_wrap(~perp)
```

```
## Warning: Removed 864641 rows containing missing values or values outside the scale range
## (`geom_col()`).
```



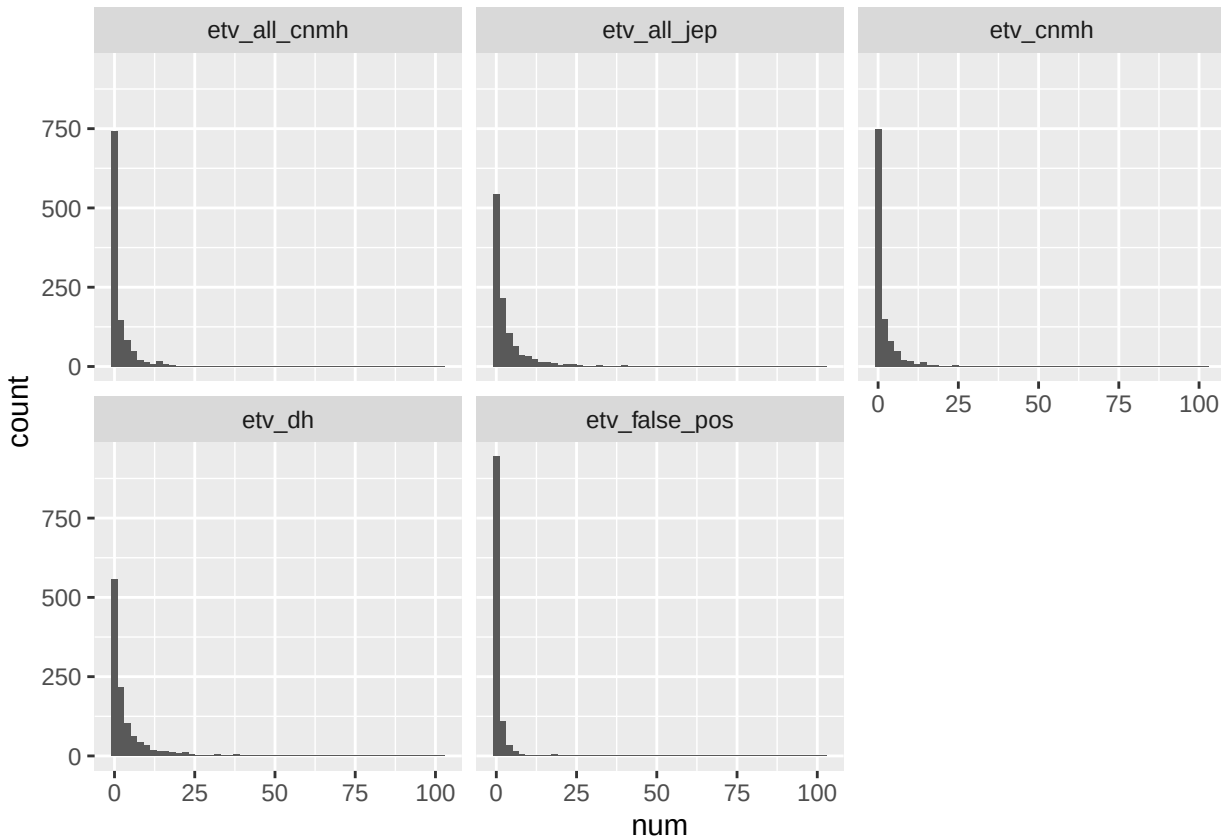

```
full_dataset |>
  dplyr::select(contains(c("year", "_fal_pos"))) |>
  group_by(year) |> #time
  pivot_longer(-year, names_to = "fal_pos", values_to = "num") |> #get column of count
  mutate(
    type = case_when( #make violence type column
      grepl("forced_disappearence", fal_pos) ~ "forced_disappearence",
      grepl("massacre", fal_pos) ~ "massacre",
      grepl("selective_assasination", fal_pos) ~ "selective_assasination",
      grepl("sexual", fal_pos) ~ "sexual"
    )
  ) |>
  ggplot(aes(x = year, y = num, fill = type)) + #plot count by year; type by color
    geom_col()
```

```
## Warning: Removed 140914 rows containing missing values or values outside the scale range
## (`geom_col()`).
```



TODO: show data with outliers in a better way…

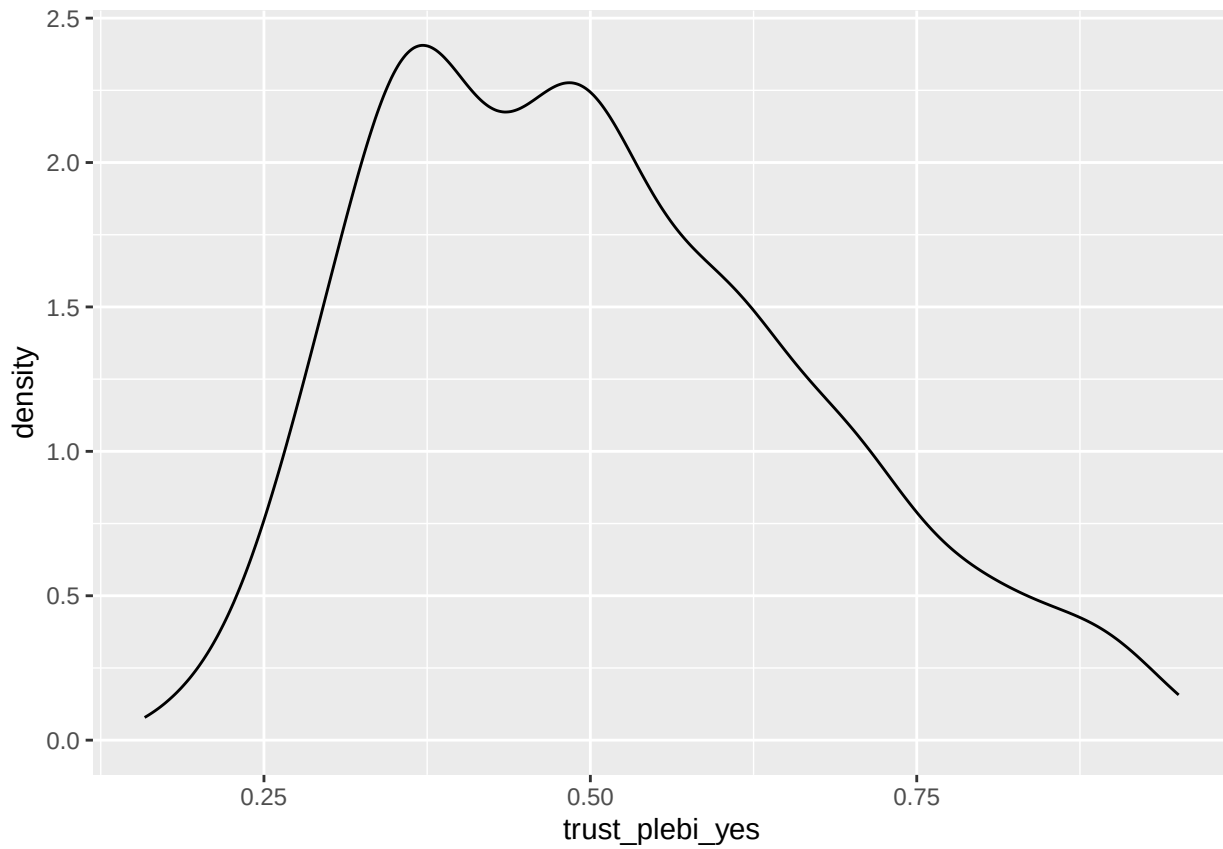
```
estimation_data |>
  dplyr::select(contains(c("_dh", "_cnmh", "_jep", "false_pos"))) |> #types
  pivot_longer(everything(), names_to = "fal_pos", values_to = "num") |> #get column of count
  ggplot(aes(x = num)) + #plot count by year; type by color
    geom_histogram(binwidth = 2) +
    facet_wrap(~fal_pos)
```



1.2 Density plot for peace support

```
estimation_data |>
  ggplot(aes(x = trust_plebi_yes)) +
    geom_density()
```

```
## Warning: Removed 2 rows containing non-finite outside the scale range
## (`stat_density()`).
```




```
rm(estimation_data, full_dataset)
```

2. Naive regression

Estimated treatment effect:

```
#naive <- lm_robust(trust_plebi_yes ~ treat, data = estimation_nona) #continuous
lm_naive <- lm_robust(trust_plebi_yes ~ treat_dicho, data = estimation_nona) #dichotomized
```

3. Naive + Control regression

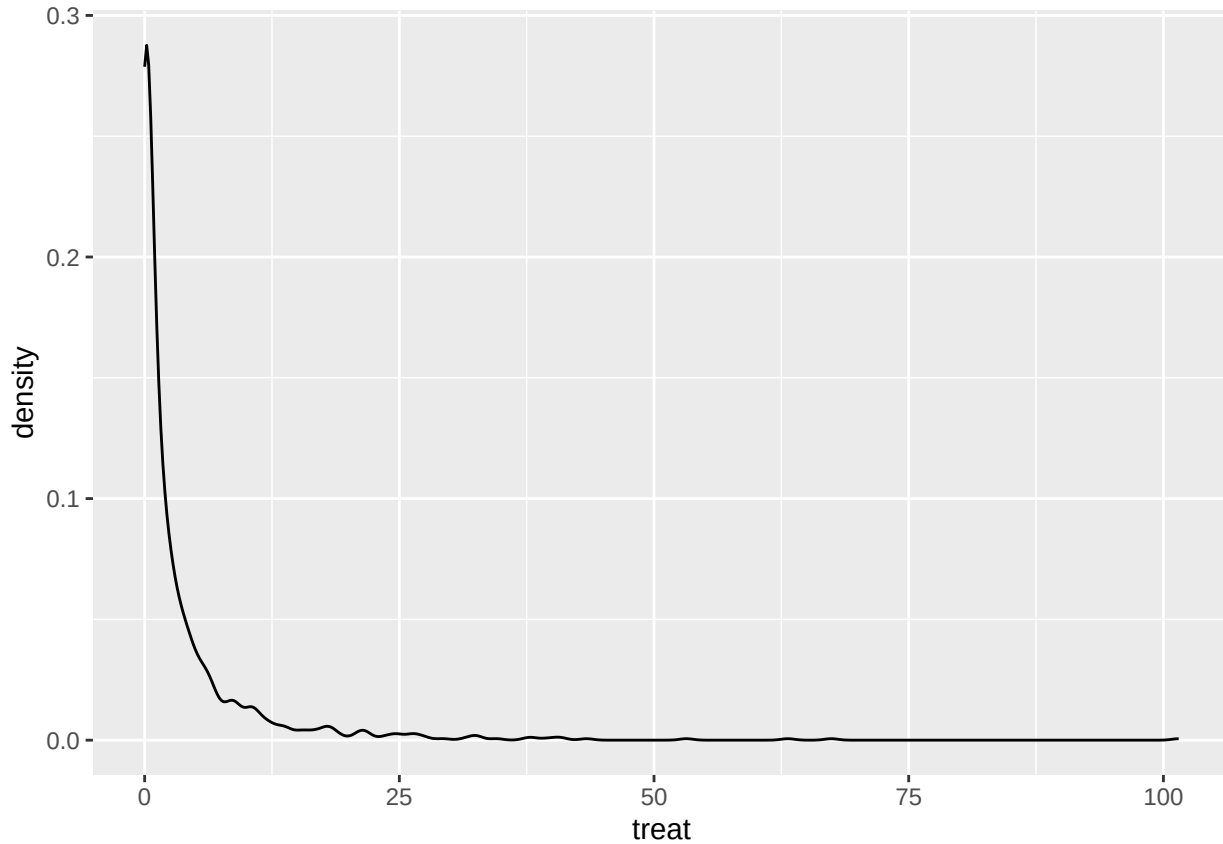
```
#lm_robust(trust_plebi_yes ~ treat + uribe_2002_proportion + uribe_2006_proportion + avg_schooling + li

lm_controls <- lm_robust(trust_plebi_yes ~ treat_dicho + uribe_2002_proportion + uribe_2006_proportion +
```

TODO: fixed effects or clustered standard errors?

4. Matching

4.1 Covariates

```
estimation_nona |>
  ggplot(aes(x = treat)) + #spread of etv proportion
    geom_density()
```




```
table(estimation_nona$treat)[1] #number of zero state violence
```

```
##   0
## 365
```

```
table(estimation_nona$treat)[[1]]/nrow(estimation_nona) #percent of municipalities that have not experi
```

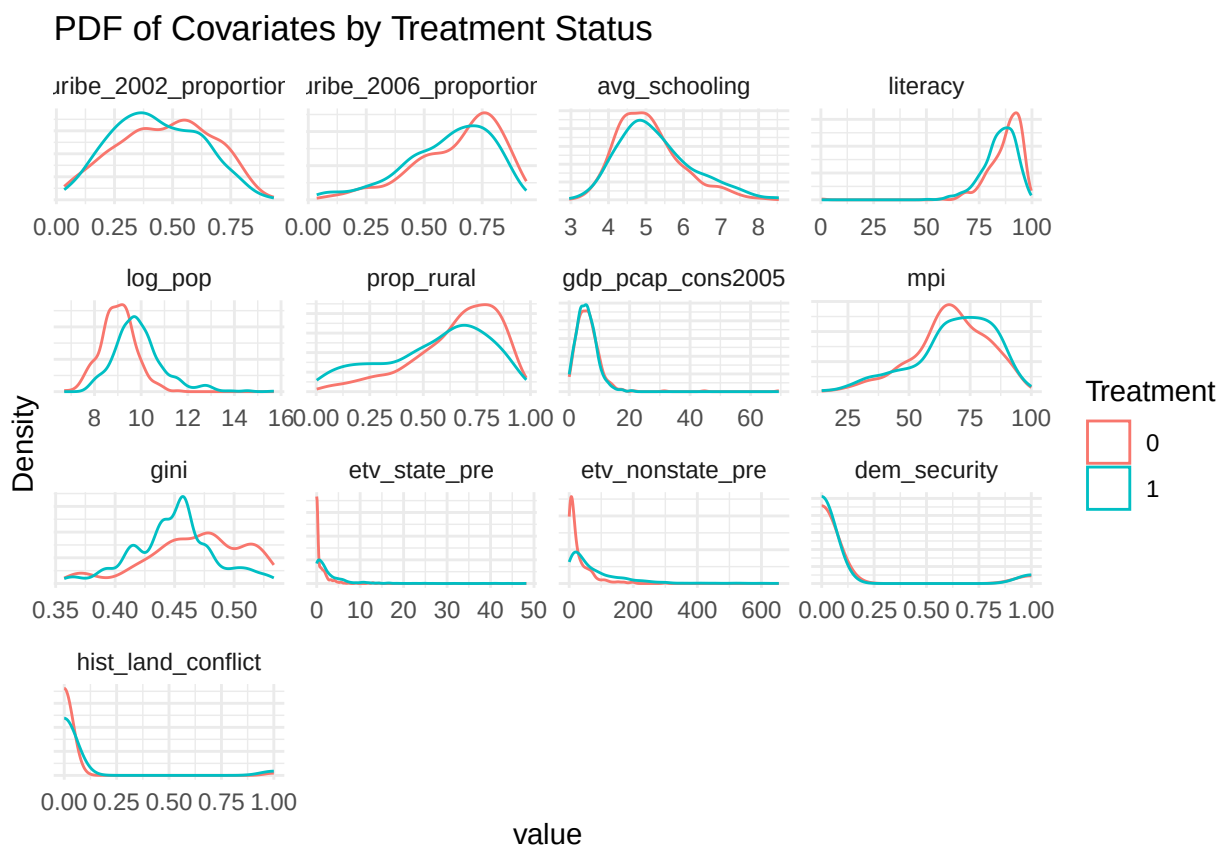
```
## [1] 0.3628231
```

4.1.1 Covariate balance

```
estimation_nona |>
  dplyr::select(all_of(covariates), treat_dicho) |>
  reshape2::melt(all_of(covariates), id.vars = "treat_dicho", na.rm = T) |>
  ggplot(aes(x = value)) +
  geom_density(aes(col = as_factor(treat_dicho)), alpha = 0.4) +
  facet_wrap(~ variable, scales = "free") +
  labs(title = paste0("PDF of Covariates by Treatment Status"),
       y = "Density", col = "Treatment") +
  labs(col = "Treatment") +
  theme_minimal() +
  theme(axis.text.y = element_blank(), axis.ticks.y = element_blank())
```

```
## Warning: Using `all_of()` outside of a selecting function was deprecated in tidyselect
## 1.2.0.
## i See details at
##   <https://tidyselect.r-lib.org/reference/faq-selection-context.html>
```

```
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```




```r
match_info <- MatchBalance(balance_formula, data = estimation_nona, nboots = 1000, print.level = 0)

balance_test_before <- baltest.collect(match_info, var.names = covariates, after = FALSE) # balance bef
balance_test_before <- balance_test_before[, c("mean.Tr", "mean.Co", "T pval", "KS pval")]

balance_test_before |>
  kable(booktabs = T, digits = 3, align = rep('c', 4),
        col.names = c("Treatment Mean", "Control Mean", "T p-value", "KS p-value")) |>
  kable_styling(full_width = F, position = "center")
```

	Treatment Mean	Control Mean	T p-value	KS p-value
uribe_2002_proportion	0.431	0.467	0.005	0.004
uribe_2006_proportion	0.603	0.649	0.000	0.000
avg_schooling	5.261	5.072	0.002	0.015
literacy	84.635	87.495	0.000	0.000
log_pop	9.896	9.001	0.000	0.000
prop_rural	0.553	0.658	0.000	0.000
gdp_pcap_cons2005	5.805	6.388	0.057	0.370
mpi	68.743	67.150	0.120	0.002
gini	0.449	0.468	0.000	0.000
etv_state_pre	3.153	0.911	0.000	0.000

etv_nonstate_pre	80.892	29.604	0.000	0.000
dem_security	0.090	0.090	0.997	NA
hist_land_conflict	0.069	0.025	0.001	NA

These are almost all significantly different between treatment and control.

4.1.2 Matching covariates

Matching throws away non-matched control units

TODO: Why ATT? We are grabbing the control unit with the closest X_i What does closest mean? For Match(): normalized Euclidean distance

curse of dimensionality: the more covariates, the further Todo: should we think through each covariate and whether we really need it? Also: issue of overspecifying the model

Issue of discrepancy with multiple continuous variables Have to adjust for it (bias correction) through BiasAdjust = TRUE Subtracts off estimated discrepancy

TODO: Different combinations: replacement vs without (of controls to match to treated); how many matches; which algorithm (Coarse - too many covariates means too far distance; pre-processing - match then regression on only close); whether to drop treatment units (removes bias)

caliper: removes units with no common support

```
exact_matches <- ifelse(colnames(estimation_nona[,covariates]) %in% c("dem_security", "hist_land_confli

match_model_1 <- Match(Y = estimation_nona$trust_plebi_yes,
                       Tr = estimation_nona$treat_dicho,
                       X = estimation_nona[, covariates],
                       M = 1, # one-to-one matching
                       exact = exact_matches,
                       caliper = 2, #0.25 standard deviations dropped
                       Weight = 2, #inverse of variance
                       replace = TRUE,
                       estimand = "ATT",
                       BiasAdjust = TRUE) # bias adjustment

match_data_1 <- estimation_nona[c(match_model_1$index.treated,
                                  match_model_1$index.control),] #596 treated, 596 control
table(match_data_1$treat_dicho)
```

```
##
##   0   1
## 597 597
```

```
# #run again - doesn't make sense
# match_model_2 <- Match(Y = match_data_1$trust_plebi_yes,
#                        Tr = match_data_1$treat_dicho,
#                        X = match_data_1[, covariates],
#                        M = 1, # one-to-one matching
#                        estimand = "ATT",
#                        BiasAdjust = TRUE) # bias adjustment
#
# match_data_2 <- estimation_nona[c(match_model_2$index.treated,
#                                   match_model_2$index.control),]
#
```

```r
# match_data_2 <- match_data_2 |>
#   drop_na()
```

```r
match_model_1$est
```

```
##              [,1]
## [1,] -0.009317198
```

```r
match_model_1$se
```

```
## [1] 0.007787961
```

```r
summary(match_model_1)
```

```
##
## Estimate...  -0.0093172
## AI SE......  0.007788
## T-stat.....  -1.1964
## p.val......  0.23156
##
## Original number of observations..............  1006
## Original number of treated obs...............  641
## Matched number of observations...............  597
## Matched number of observations  (unweighted).  597
##
## Number of obs dropped by 'exact' or 'caliper'  44
```

```r
# Estimate...  -0.011164
# AI SE......  0.0077703
# T-stat.....  -1.4367
# p.val......  0.1508

#calculate ATT manually
mean(match_data_1$trust_plebi_yes[match_data_1$treat_dicho == 1]) - mean(match_data_1$trust_plebi_yes[ma
```

```
## [1] 0.005771745
```

```r
#naive matching
lm_match <- lm_robust(trust_plebi_yes ~ treat_dicho, data = match_data_1) #0.005162898

#controls matching
lm_match_controls <- lm_robust(trust_plebi_yes ~ treat_dicho + uribe_2002_proportion + uribe_2006_propo
```

Checking the balance after the matching

```r
match_info <- MatchBalance(match.out = match_model_1,
                                   formul = balance_formula,
                                   data = estimation_nona,
                                   print.level = 0, nboots = 1000)

balance_test_after <- baltest.collect(match_info, var.names = covariates, after = TRUE)
balance_test_after <- balance_test_after[, c("mean.Tr", "mean.Co", "T pval", "KS pval")]

balance_test_after %>%
  kable(booktabs = T, digits = 3, align = rep('c', 4),
        col.names = c("Treatment Mean", "Control Mean", "T p-value", "KS p-value")) %>%
  kable_styling(full_width = F, position = "center")
```

	Treatment Mean	Control Mean	T p-value	KS p-value
uribe_2002_proportion	0.430	0.457	0.000	0.000
uribe_2006_proportion	0.606	0.640	0.000	0.003
avg_schooling	5.262	5.153	0.000	0.012
literacy	84.847	86.223	0.000	0.001
log_pop	9.846	9.444	0.000	0.000
prop_rural	0.556	0.594	0.000	0.009
gdp_pcap_cons2005	5.771	6.046	0.004	0.001
mpi	68.747	68.256	0.184	0.305
gini	0.449	0.455	0.000	0.000
etv_state_pre	2.480	1.391	0.000	0.000
etv_nonstate_pre	68.949	42.110	0.000	0.000
dem_security	0.074	0.074	1.000	NA
hist_land_conflict	0.059	0.059	1.000	NA

```
# #second round - the same
# match_info_1 <- MatchBalance(match.out = match_model_2,
#                                     formul = balance_formula,
#                                     data = match_data_1,
#                                     print.level = 0, nboots = 1000)
#
# match_info_1 <- MatchBalance(balance_formula,
#                                     data = match_data_2,
#                                     print.level = 0, nboots = 1000)
#
# balance_test_after_1 <- baltest.collect(match_info_1, var.names = covariates, after = FALSE)
# balance_test_after_1 <- balance_test_after[, c("mean.Tr", "mean.Co", "T pval", "KS pval")]
# balance_test_after_1 %>%
#   kable(booktabs = T, digits = 3, align = rep('c', 4),
#         col.names = c("Treatment Mean", "Control Mean", "T p-value", "KS p-value")) %>%
#   kable_styling(full_width = F, position = "center")
```

MatchIt; 1-on-1 nearest neighbor matching on the scaled euclidian distance, with replacement

```
# https://kosukeimai.github.io/MatchIt/articles/matching-methods.html
# https://kosukeimai.github.io/MatchIt/articles/estimating-effects.html#the-standard-case


match_model_2 <- matchit(balance_formula,
                         data = estimation_nona,
                         estimand = "ATT",
                         method = "nearest",
                         ratio = 1,
                         distance = "scaled_euclidean",
                         replace = TRUE)

summary(match_model_2)
```

```
##
## Call:
## matchit(formula = balance_formula, data = estimation_nona, method = "nearest",
##     distance = "scaled_euclidean", estimand = "ATT", replace = TRUE,
```

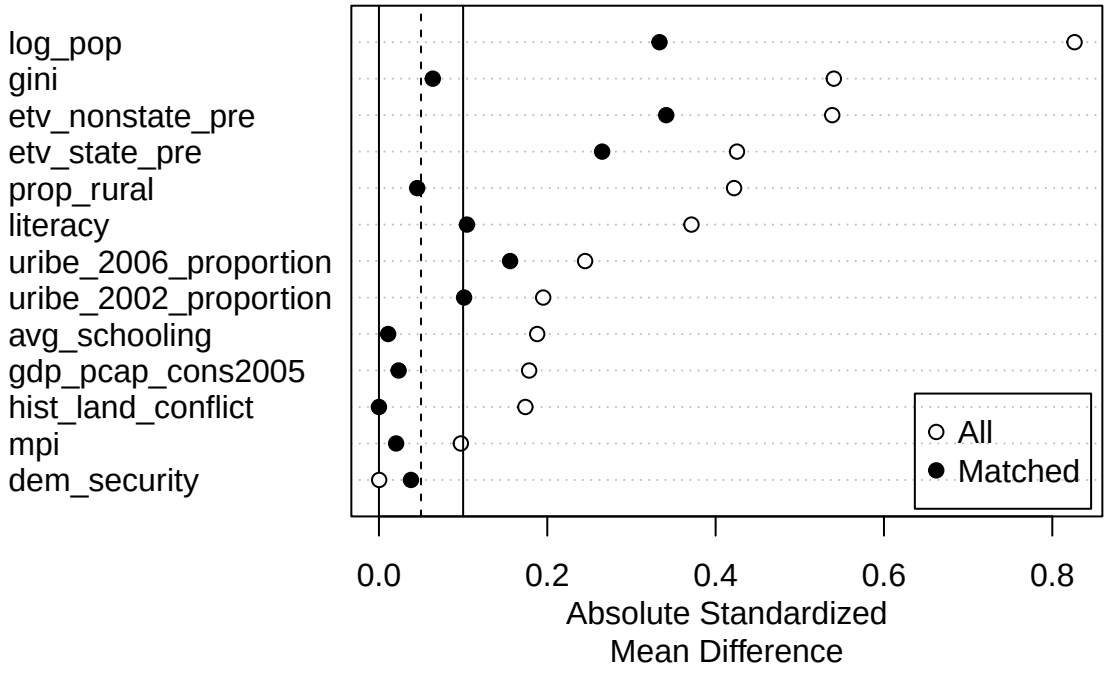
```
##     ratio = 1)
##
## Summary of Balance for All Data:
##                        Means Treated Means Control Std. Mean Diff. Var. Ratio
## uribe_2002_proportion         0.4305        0.4672         -0.1951     0.8725
## uribe_2006_proportion         0.6027        0.6494         -0.2449     1.0747
## avg_schooling                 5.2606        5.0724          0.1880     1.3631
## literacy                     84.6350       87.4946         -0.3713     0.8288
## log_pop                       9.8961        9.0006          0.8264     2.1545
## prop_rural                    0.5532        0.6579         -0.4220     1.5322
## gdp_pcap_cons2005             5.8053        6.3880         -0.1784     0.3800
## mpi                          68.7435       67.1497          0.0973     1.1685
## gini                          0.4491        0.4677         -0.5404     0.7056
## etv_state_pre                 3.1533        0.9108          0.4254     8.0147
## etv_nonstate_pre             80.8921       29.6035          0.5385     5.9304
## dem_security                  0.0905        0.0904          0.0003          .
## hist_land_conflict            0.0686        0.0247          0.1740          .
##                        eCDF Mean eCDF Max
## uribe_2002_proportion     0.0575   0.1133
## uribe_2006_proportion     0.0764   0.1470
## avg_schooling             0.0524   0.1015
## literacy                  0.1263   0.2178
## log_pop                   0.2592   0.4189
## prop_rural                0.1243   0.1932
## gdp_pcap_cons2005         0.0222   0.0590
## mpi                       0.0480   0.1109
## gini                      0.1395   0.3106
## etv_state_pre             0.2521   0.4233
## etv_nonstate_pre          0.2284   0.3353
## dem_security              0.0001   0.0001
## hist_land_conflict        0.0440   0.0440
##
## Summary of Balance for Matched Data:
##                        Means Treated Means Control Std. Mean Diff. Var. Ratio
## uribe_2002_proportion         0.4305        0.4495         -0.1011     1.0026
## uribe_2006_proportion         0.6027        0.6324         -0.1558     1.1185
## avg_schooling                 5.2606        5.2497          0.0109     1.1738
## literacy                     84.6350       85.4403         -0.1045     1.1951
## log_pop                       9.8961        9.5349          0.3333     1.6958
## prop_rural                    0.5532        0.5644         -0.0454     1.2196
## gdp_pcap_cons2005             5.8053        5.8815         -0.0233     1.0710
## mpi                          68.7435       68.4089          0.0204     0.9432
## gini                          0.4491        0.4513         -0.0640     1.0574
## etv_state_pre                 3.1533        1.7553          0.2652     3.1598
## etv_nonstate_pre             80.8921       48.3866          0.3413     2.4870
## dem_security                  0.0905        0.0796          0.0381          .
## hist_land_conflict            0.0686        0.0686          0.0000          .
##                        eCDF Mean eCDF Max Std. Pair Dist.
## uribe_2002_proportion     0.0310   0.1170          0.4898
## uribe_2006_proportion     0.0489   0.0936          0.4393
## avg_schooling             0.0224   0.0686          0.4074
## literacy                  0.0290   0.0842          0.4901
## log_pop                   0.0866   0.1825          0.5373
## prop_rural                0.0195   0.0671          0.4274
```

```
## gdp_pcap_cons2005        0.0218   0.0780          0.5751
## mpi                      0.0105   0.0437          0.4038
## gini                     0.0262   0.0998          0.4450
## etv_state_pre            0.1422   0.2106          0.3757
## etv_nonstate_pre         0.1247   0.2012          0.4438
## dem_security             0.0109   0.0109          0.0381
## hist_land_conflict       0.0000   0.0000          0.0000
##
## Sample Sizes:
##                Control Treated
## All              365.     641
## Matched (ESS)     89.5    641
## Matched          208.     641
## Unmatched        157.       0
## Discarded          0.       0
```

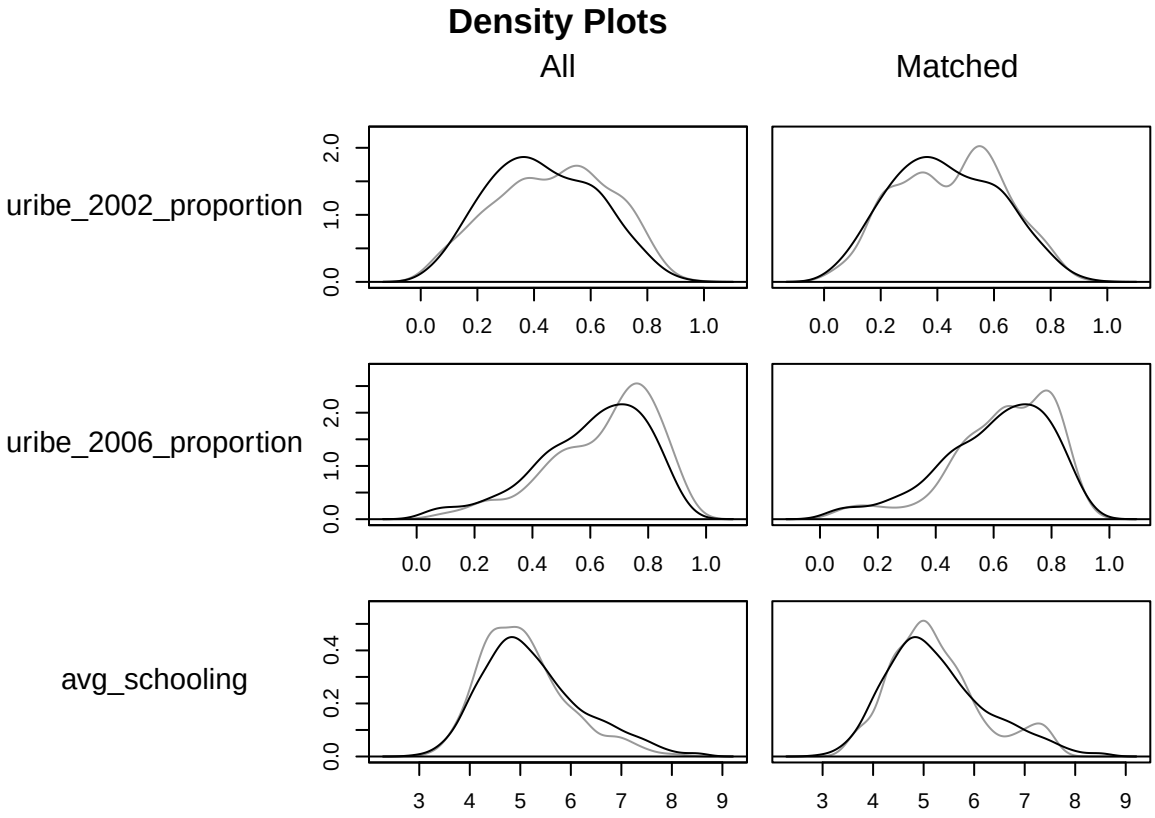
```r
match_data_2 <- match_data(match_model_2) # 849 obs
table(match_data_2$treat_dicho) #640 treated, 209 untreated
```

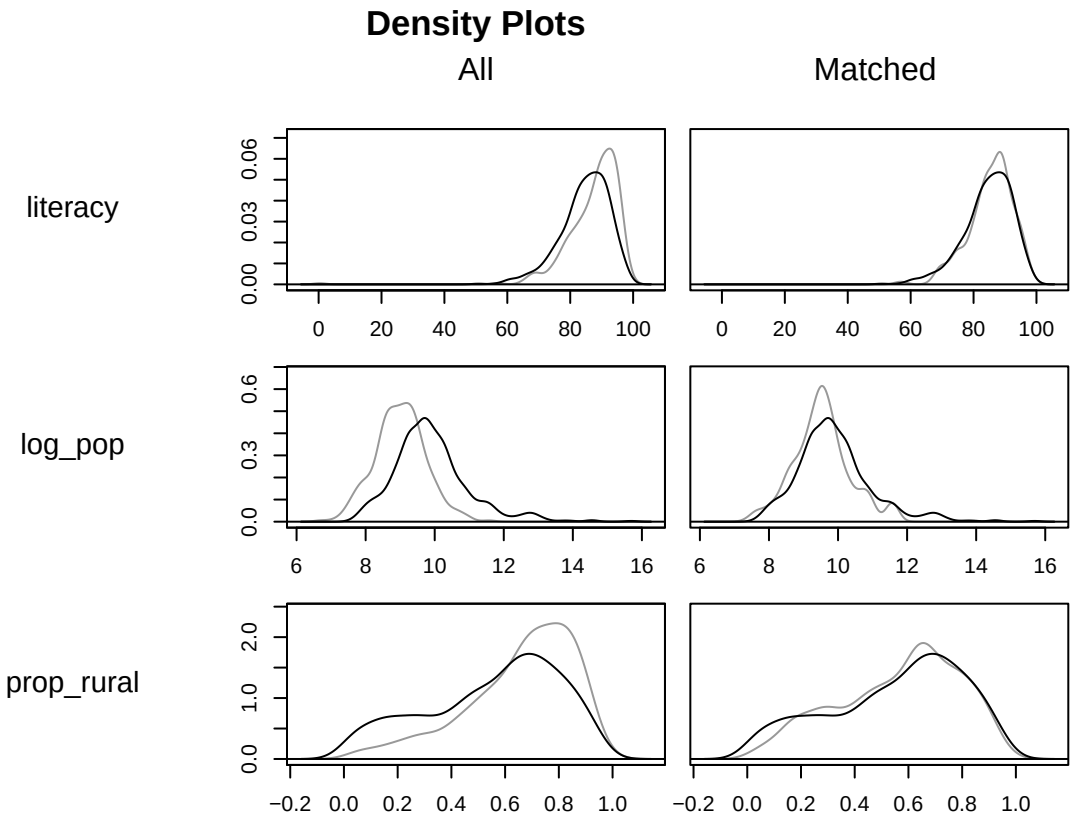
```
##
##   0   1
## 208 641
```

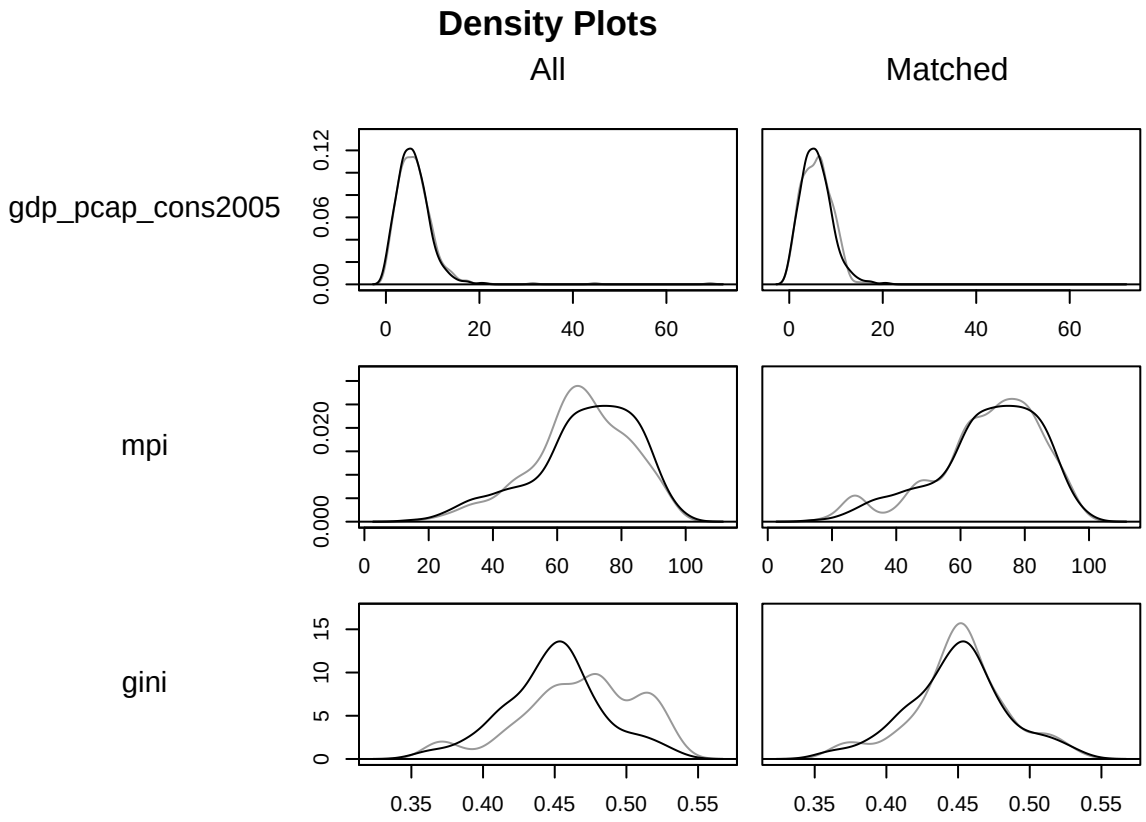
```r
plot(summary(match_model_2), var.order = "unmatched")
```

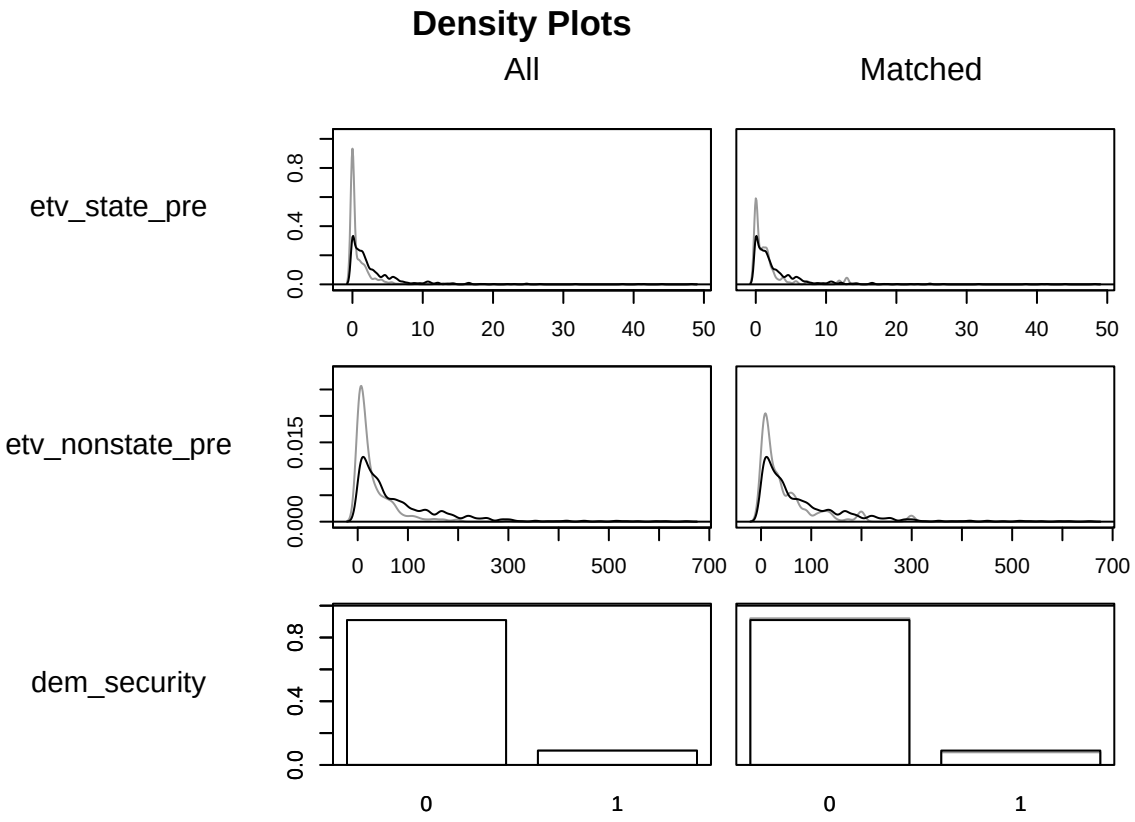


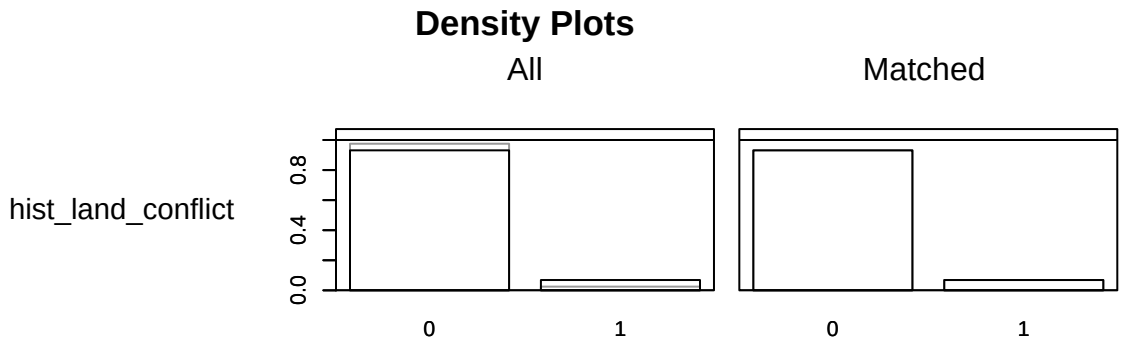
```r
plot(match_model_2, type = "density", interactive = FALSE)
```




Density Plots
All
Matched
literacy
log_pop
prop_rural
0.00
0.03
0.06
0
20
40
60
80
100
0.0
0.3
0.6
6
8
10
12
14
16
0.0
1.0
2.0
−0.2
0.0
0.2
0.4
0.6
0.8
1.0


Density Plots
All
Matched
gdp_pcap_cons2005
mpi
gini
0.00
0.06
0.12
0
20
40
60
0.000
0.020
80
100
0
5
10
15
0.35
0.40
0.45
0.50
0.55


Density Plots
All
Matched
etv_state_pre
etv_nonstate_pre
dem_security
0.0
0.4
0.8
0
10
20
30
40
50
0.000
0.015
0
100
300
500
700
0
1



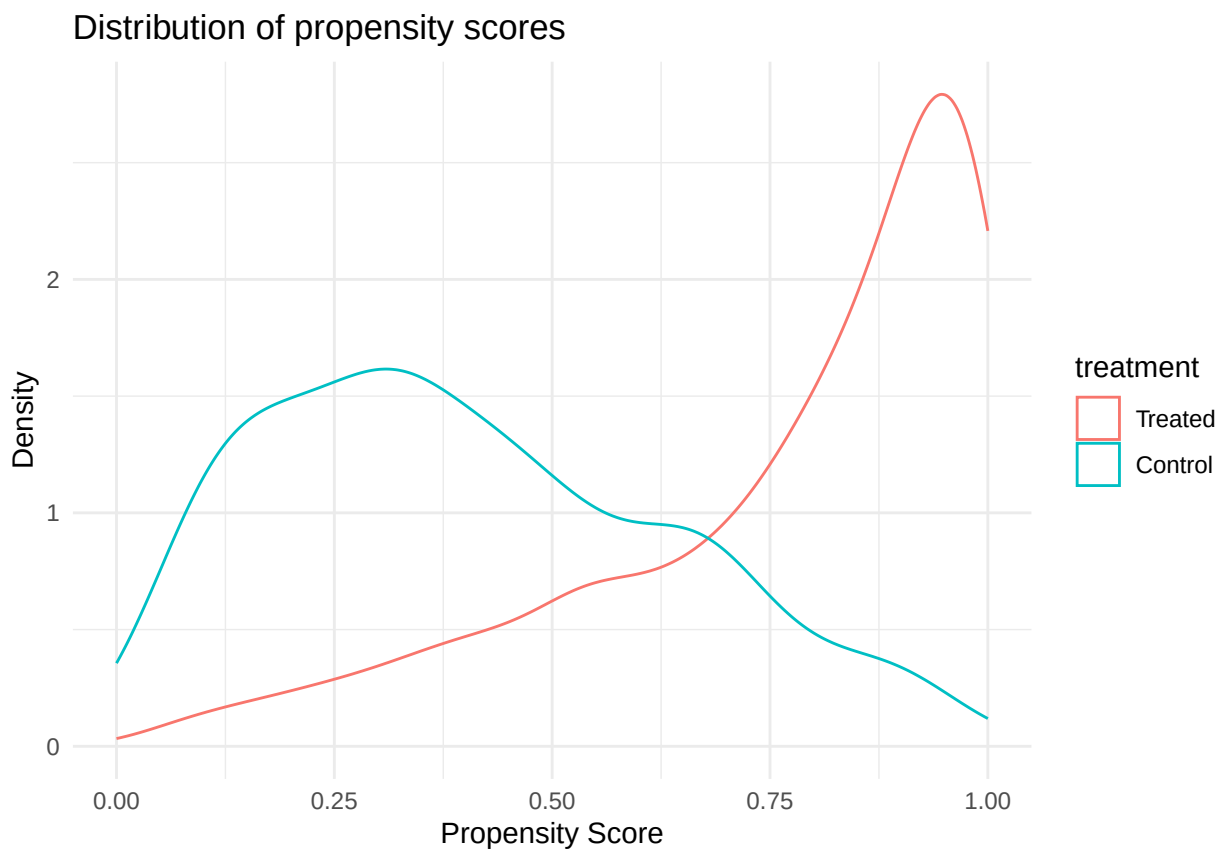

```
lm_match2 <- lm_robust(trust_plebi_yes ~ treat_dicho, weights = weights, data = match_data_2) #-0.00099

lm_match_controls2 <- lm_robust(trust_plebi_yes ~ treat_dicho + uribe_2002_proportion + uribe_2006_prop

#use cobalt: https://kosukeimai.github.io/MatchIt/articles/assessing-balance.html#assessing-balance-wit
```

```
rm(balance_test_before, balance_test_after, match_info, match_data_1, match_data_2, match_model_1, matc
```

4.2 Propensity score matching

4.2.1 Propensity score balance

```
# plot
ggplot(tibble(ps = estimation_nona$ps,
              treatment = as_factor(ifelse(estimation_nona$treat_dicho == 1, "Treated", "Control"))),
       aes(x = ps, col = treatment)) +
  geom_density() +
  labs(title = "Distribution of propensity scores", y = "Density", x = "Propensity Score") +
  xlim(c(0, 1)) +
  theme_minimal()
```



Very different; match by propensity to get balanced units.

4.2.2 Matching propensity scores

Compare units that have a comparable probability of getting treated.

```
match_ps <- Match(Y = estimation_nona$trust_plebi_yes,
                  Tr = estimation_nona$treat_dicho,
                  X = estimation_nona$ps,
                  M = 1,
                  estimand = "ATT")

match_data_ps <- estimation_nona[c(match_ps$index.treated,
                                   match_ps$index.control),] #723 treated, 723 control
table(match_data_ps$treat_dicho)
```

```
##
##   0   1
## 731 731
```

```
match_ps$est
```

```
##            [,1]
## [1,] 0.05284852
```

```
match_ps$se
```

```
## [1] 0.03219991
```

```r
summary(match_ps)
```

```
##
## Estimate...  0.052849
## AI SE......  0.0322
## T-stat.....  1.6413
## p.val......  0.10074
##
## Original number of observations..............  1006
## Original number of treated obs...............  641
## Matched number of observations...............  641
## Matched number of observations  (unweighted).  731
```

```r
# Estimate...  0.045338
# AI SE......  0.032178
# T-stat.....  1.409
# p.val......  0.15885

#calculate ATT manually
mean(match_data_ps$trust_plebi_yes[match_data_ps$treat_dicho == 1]) - mean(match_data_ps$trust_plebi_yes
```

```
## [1] 0.0399034
```

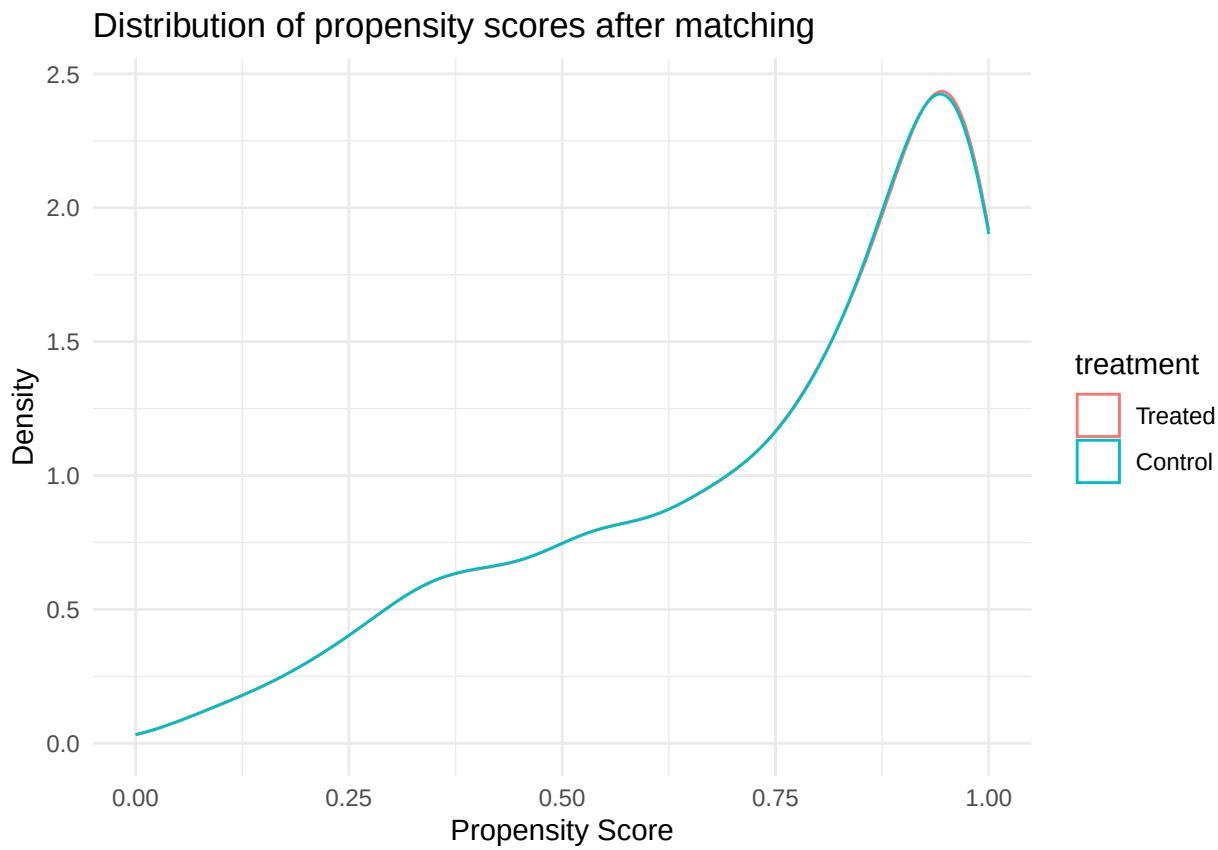
```r
#naive matching
lm_match_ps <- lm_robust(trust_plebi_yes ~ treat_dicho, data = match_data_ps) #0.03622134

#controls matching
lm_match_ps_controls <- lm_robust(trust_plebi_yes ~ treat_dicho + uribe_2002_proportion + uribe_2006_pro
```

```r
match_balance_ps <- MatchBalance(match.out = match_ps,
                                 formul = balance_formula,
                                 data = estimation_nona,
                                 print.level = 0)

balance_table_ps <- baltest.collect(match_balance_ps, var.names = covariates, after = TRUE)
balance_table_ps <- balance_table_ps[, c("mean.Tr", "mean.Co", "T pval", "KS pval")]

balance_table_ps |>
  kable(booktabs = T, digits = 3, align = rep('c', 4),
        col.names = c("Treatment Mean", "Control Mean", "T p-value", "KS p-value")) |>
  kable_styling(full_width = F, position = "center")
```

	Treatment Mean	Control Mean	T p-value	KS p-value
uribe_2002_proportion	0.431	0.518	0.000	0
uribe_2006_proportion	0.603	0.693	0.000	0
avg_schooling	5.261	5.008	0.000	0
literacy	84.635	80.316	0.000	0
log_pop	9.896	9.638	0.000	0
prop_rural	0.553	0.596	0.001	0
gdp_pcap_cons2005	5.805	5.706	0.581	0
mpi	68.743	71.524	0.002	0
gini	0.449	0.455	0.003	0
etv_state_pre	3.153	3.224	0.729	0
etv_nonstate_pre	80.892	96.496	0.000	0

dem_security	0.090	0.083	0.629	NA
hist_land_conflict	0.069	0.014	0.000	NA

```
ggplot(tibble(ps = match_data_ps$ps,
              treatment = as_factor(ifelse(match_data_ps$treat_dicho == 1, "Treated", "Control"))),
       aes(x = ps, col = treatment)) +
  geom_density() +
  labs(title = "Distribution of propensity scores after matching", y = "Density", x = "Propensity Score
  xlim(c(0, 1)) +
  theme_minimal()
```



MatchIt: 1-on-1 nearest neighbor matching on the propensity score, with replacement

exact = ~dem_security + hist_land_conflict,

```
match_balance_ps_2 <- matchit(balance_formula,
                     data = estimation_nona,
                     estimand = "ATT",
                     method = "nearest",
                     ratio = 1,
                     distance = "glm",
                     link = "logit",
                     replace = TRUE)

match_balance_ps_2
```

```
## A `matchit` object
```

```
##  - method: 1:1 nearest neighbor matching with replacement
##  - distance: Propensity score
##              - estimated with logistic regression
##  - number of obs.: 1006 (original), 799 (matched)
##  - target estimand: ATT
##  - covariates: uribe_2002_proportion, uribe_2006_proportion, avg_schooling, literacy, log_pop, prop_
```

```r
summary(match_balance_ps_2)
```

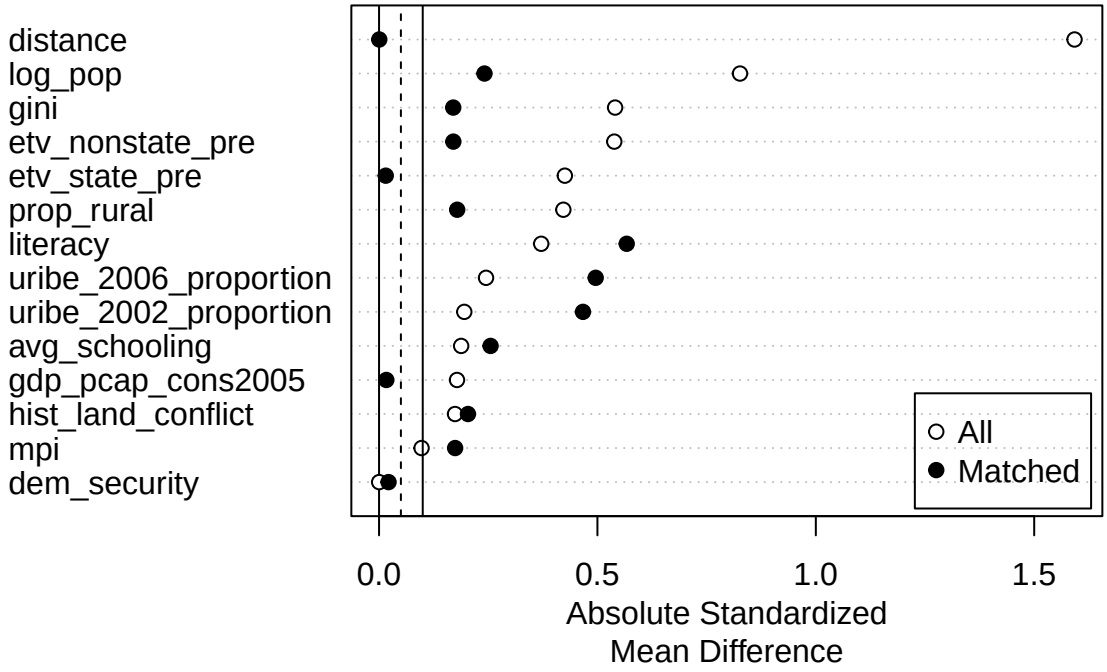
```
##
## Call:
## matchit(formula = balance_formula, data = estimation_nona, method = "nearest",
##     distance = "glm", link = "logit", estimand = "ATT", replace = TRUE,
##     ratio = 1)
##
## Summary of Balance for All Data:
##                       Means Treated Means Control Std. Mean Diff. Var. Ratio
## distance                     0.7700        0.4040          1.5924     0.9813
## uribe_2002_proportion        0.4305        0.4672         -0.1951     0.8725
## uribe_2006_proportion        0.6027        0.6494         -0.2449     1.0747
## avg_schooling                5.2606        5.0724          0.1880     1.3631
## literacy                    84.6350       87.4946         -0.3713     0.8288
## log_pop                      9.8961        9.0006          0.8264     2.1545
## prop_rural                   0.5532        0.6579         -0.4220     1.5322
## gdp_pcap_cons2005            5.8053        6.3880         -0.1784     0.3800
## mpi                         68.7435       67.1497          0.0973     1.1685
## gini                         0.4491        0.4677         -0.5404     0.7056
## etv_state_pre                3.1533        0.9108          0.4254     8.0147
## etv_nonstate_pre            80.8921       29.6035          0.5385     5.9304
## dem_security                 0.0905        0.0904          0.0003          .
## hist_land_conflict           0.0686        0.0247          0.1740          .
##                       eCDF Mean eCDF Max
## distance                 0.3602   0.5842
## uribe_2002_proportion    0.0575   0.1133
## uribe_2006_proportion    0.0764   0.1470
## avg_schooling            0.0524   0.1015
## literacy                 0.1263   0.2178
## log_pop                  0.2592   0.4189
## prop_rural               0.1243   0.1932
## gdp_pcap_cons2005        0.0222   0.0590
## mpi                      0.0480   0.1109
## gini                     0.1395   0.3106
## etv_state_pre            0.2521   0.4233
## etv_nonstate_pre         0.2284   0.3353
## dem_security             0.0001   0.0001
## hist_land_conflict       0.0440   0.0440
##
## Summary of Balance for Matched Data:
##                       Means Treated Means Control Std. Mean Diff. Var. Ratio
## distance                     0.7700        0.7698          0.0005     0.9637
## uribe_2002_proportion        0.4305        0.5182         -0.4669     0.8032
## uribe_2006_proportion        0.6027        0.6973         -0.4960     1.3512
## avg_schooling                5.2606        5.0050          0.2553     1.1806
## literacy                    84.6350       80.2677          0.5670     0.1829
## log_pop                      9.8961        9.6346          0.2413     2.0212
```

```
## prop_rural                   0.5532        0.5976         -0.1789     1.2958
## gdp_pcap_cons2005            5.8053        5.7509          0.0167     1.0557
## mpi                         68.7435       71.5971         -0.1742     1.0927
## gini                         0.4491        0.4550         -0.1699     0.8327
## etv_state_pre                3.1533        3.2341         -0.0153     1.5018
## etv_nonstate_pre            80.8921       97.0919         -0.1701     0.9499
## dem_security                 0.0905        0.0842          0.0218          .
## hist_land_conflict           0.0686        0.0172          0.2036          .
##                       eCDF Mean eCDF Max Std. Pair Dist.
## distance                 0.0074   0.0733          0.0132
## uribe_2002_proportion    0.1324   0.2933          1.3067
## uribe_2006_proportion    0.1662   0.3151          1.1202
## avg_schooling            0.0654   0.1544          1.1058
## literacy                 0.0575   0.1435          1.4154
## log_pop                  0.0504   0.1622          0.8091
## prop_rural               0.0546   0.1310          1.0967
## gdp_pcap_cons2005        0.0439   0.1061          1.0911
## mpi                      0.0571   0.1732          1.0485
## gini                     0.0536   0.1981          1.1255
## etv_state_pre            0.0400   0.1186          0.6181
## etv_nonstate_pre         0.0540   0.1342          0.7284
## dem_security             0.0062   0.0062          0.5547
## hist_land_conflict       0.0515   0.0515          0.3394
##
## Sample Sizes:
##               Control Treated
## All            365.       641
## Matched (ESS)   26.69     641
## Matched        158.       641
## Unmatched      207.         0
## Discarded        0.         0
```

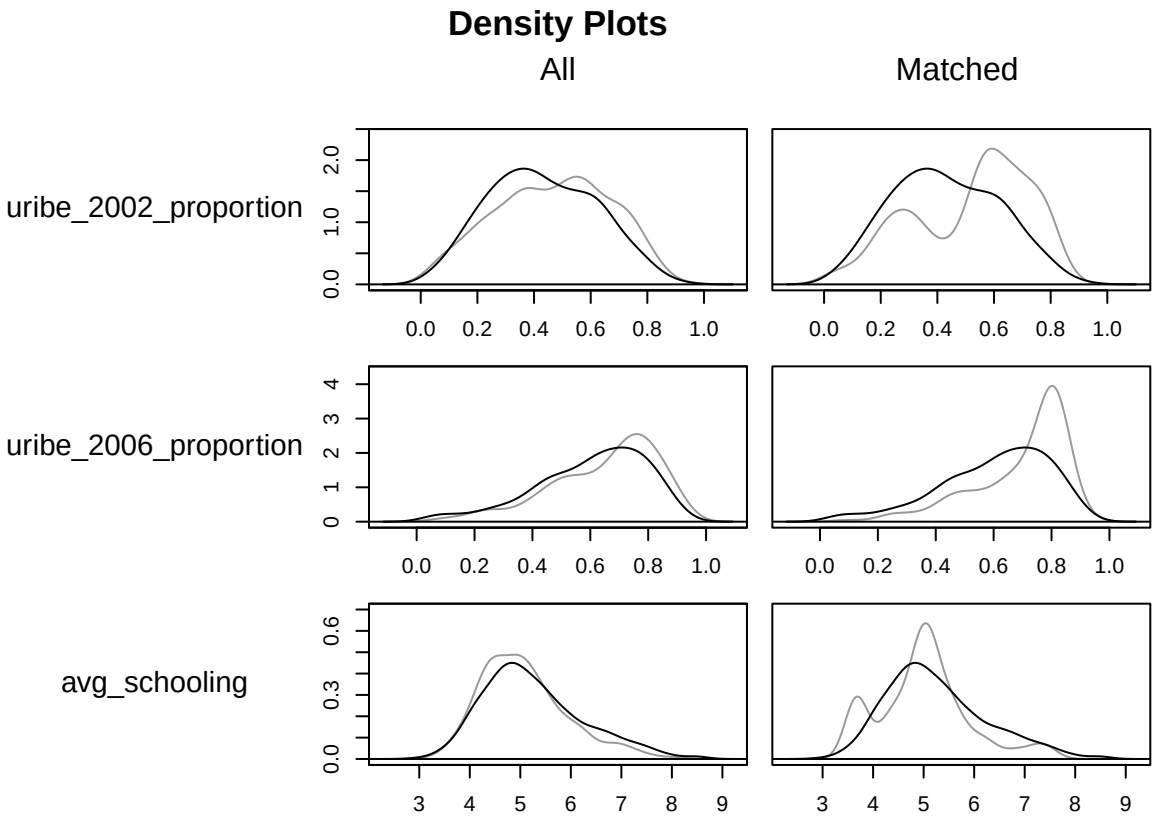
```r
match_data_ps_2 <- match_data(match_balance_ps_2) #794 obs
table(match_data_ps_2$treat_dicho) #640 treated, 154 untreated
```

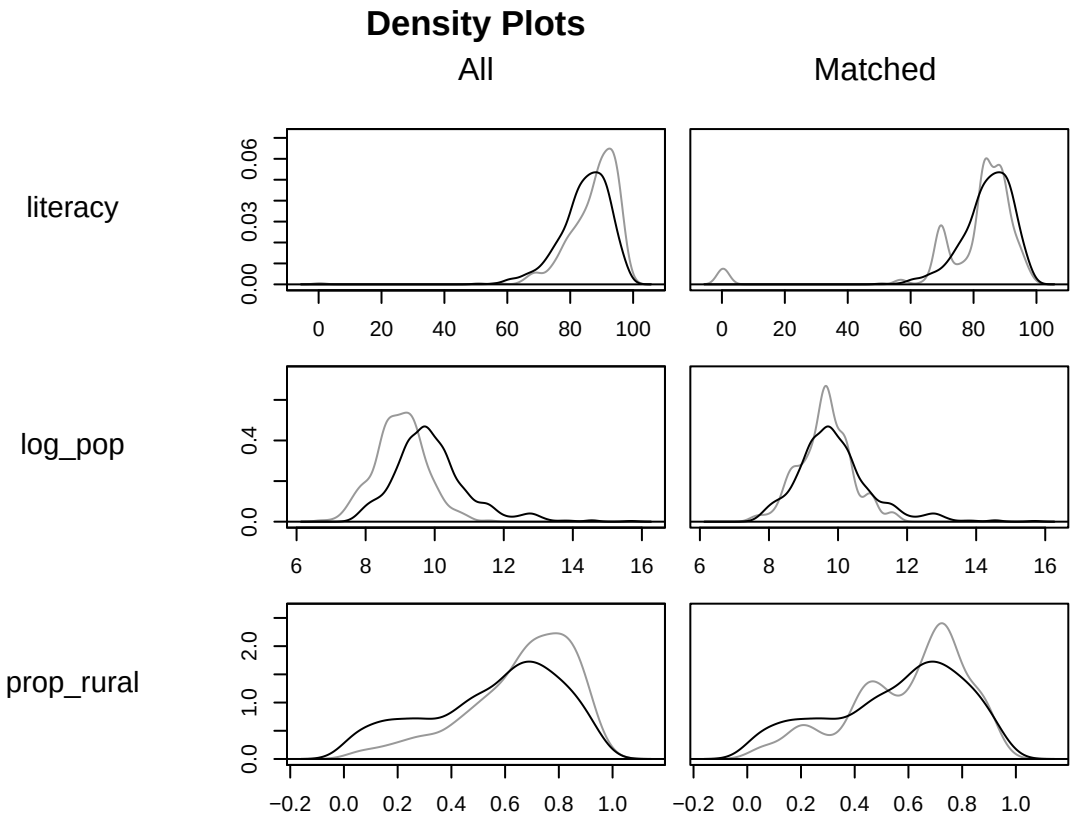
```
##
##   0   1
## 158 641
```

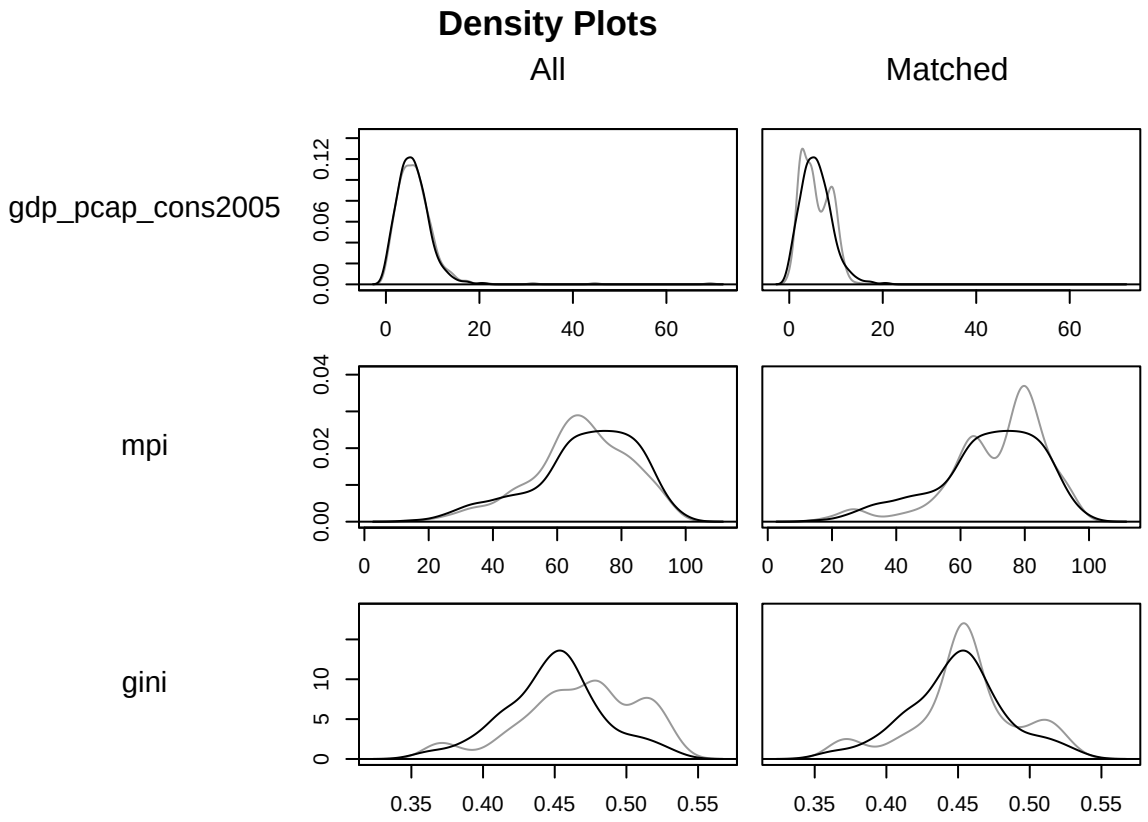
```r
plot(summary(match_balance_ps_2), var.order = "unmatched")
```

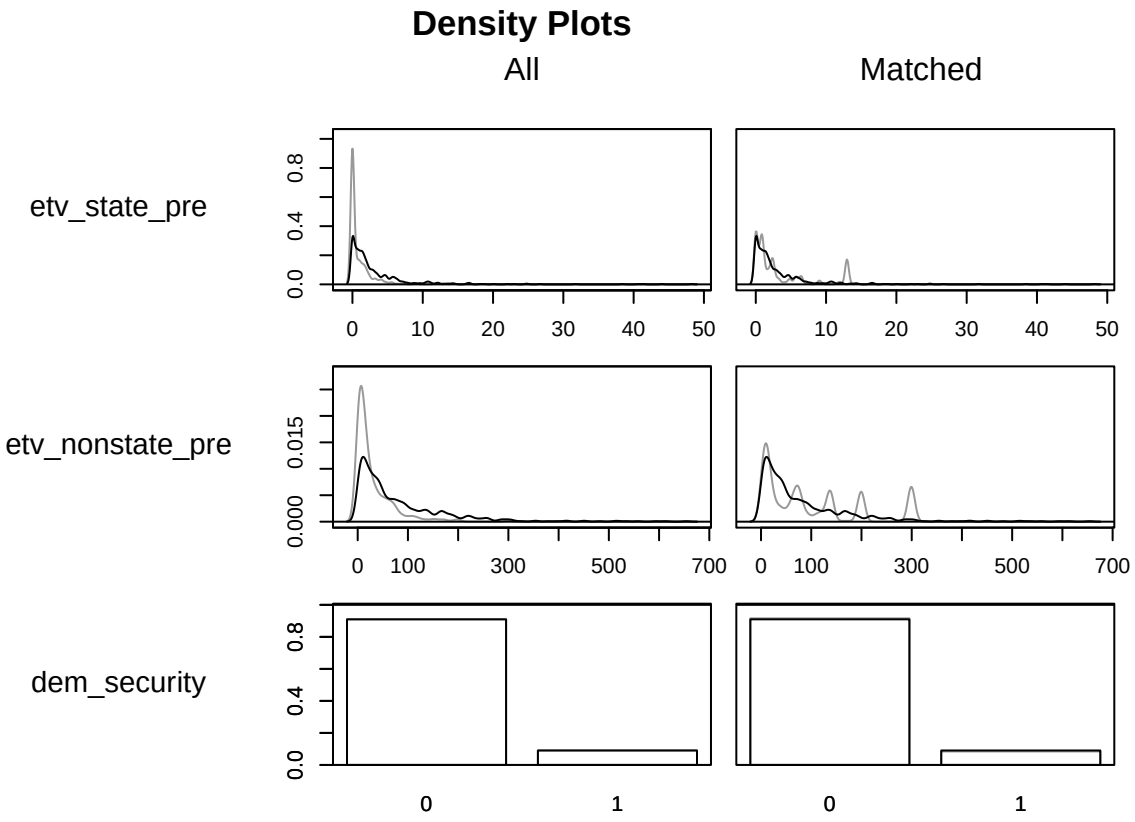



```
plot(match_balance_ps_2, type = "density", interactive = FALSE)
```


Density Plots
All
Matched
uribe_2002_proportion
uribe_2006_proportion
avg_schooling
0.0
1.0
2.0
0 1 2 3 4
0.0 0.3 0.6
0.0 0.2 0.4 0.6 0.8 1.0
3 4 5 6 7 8 9


Density Plots
All
Matched
literacy
log_pop
prop_rural
0.00
0.03
0.06
0
20
40
60
80
100
0.0
0.4
6
8
10
12
14
16
0.0
1.0
2.0
−0.2
0.0
0.2
0.4
0.6
0.8
1.0


Density Plots
All
Matched
gdp_pcap_cons2005
mpi
gini
0.00
0.06
0.12
0
20
40
60
0.00
0.02
0.04
0
20
40
60
80
100
0
5
10
0.35
0.40
0.45
0.50
0.55


Density Plots
All
Matched
etv_state_pre
etv_nonstate_pre
dem_security
0.0
0.4
0.8
0
10
20
30
40
50
0.000
0.015
100
300
500
700
1

```
lm_match_ps2 <- lm_robust(trust_plebi_yes ~ treat_dicho, weights = weights, data = match_data_ps_2) #0.

lm_match_ps_controls2 <- lm_robust(trust_plebi_yes ~ treat_dicho + uribe_2002_proportion + uribe_2006_p

ggplot(tibble(ps = match_data_ps_2$ps,
              treatment = as_factor(ifelse(match_data_ps_2$treat_dicho == 1, "Treated", "Control"))),
       aes(x = ps, col = treatment)) +
  geom_density() +
  labs(title = "Distribution of propensity scores after matching", y = "Density", x = "Propensity Score
  xlim(c(0, 1)) +
  theme_minimal()
```

Some matched are worse than all Matching generally looks better with cnmh than jep

5. Weighting

5.1 Weighting propensity scores

Inverse Probability-of-Treatment Weighting (IPW)

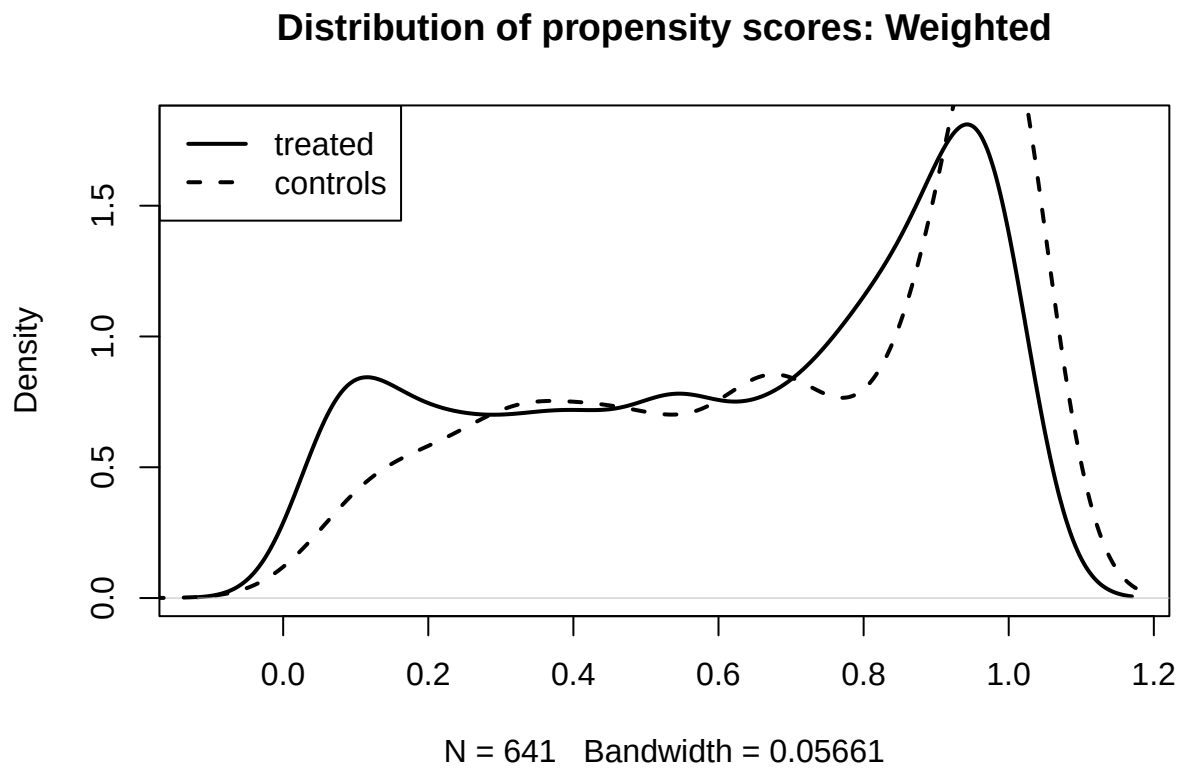
```
# stabilized IPW weights
ps = estimation_nona$ps
D = estimation_nona$treat_dicho
PrD = mean(D)
IPW = (D*PrD+(1-D)*(1-PrD))/(D*ps+(1-D)*(1-ps))

# check the propensity scores after weighting
plot(density(ps[D==1], weight = IPW[D==1]/sum(IPW[D==1])), lwd = 2,
     main="Distribution of propensity scores: Weighted")
```

```
## Warning in density.default(ps[D == 1], weight = IPW[D == 1]/sum(IPW[D == :
## Selecting bandwidth *not* using 'weights'
```

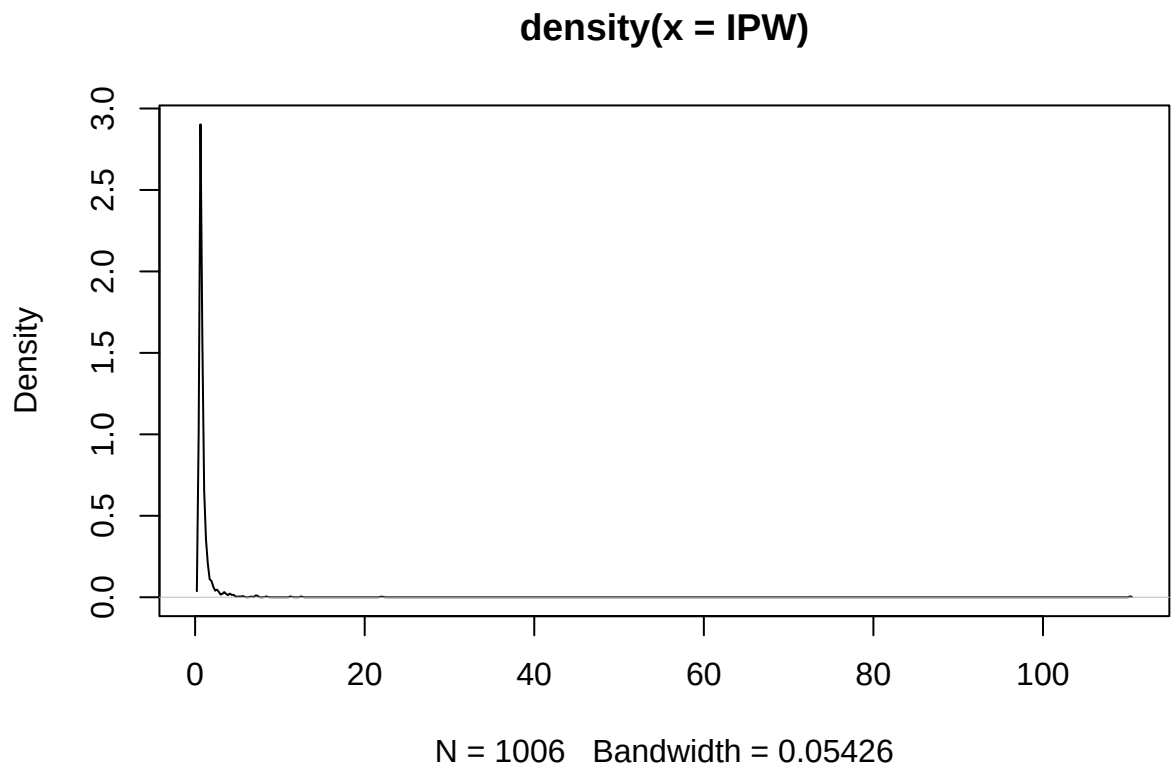
```
lines(density(ps[D==0], weight = IPW[D==0]/sum(IPW[D==0])),lwd = 2, lty = 2)
```

```
## Warning in density.default(ps[D == 0], weight = IPW[D == 0]/sum(IPW[D == :
## Selecting bandwidth *not* using 'weights'
```

```
legend("topleft", legend = c("treated","controls"), lty = c(1, 2), lwd = 2)
```




```r
plot(density(IPW)) #check IPW for extremes
```




```
#naive matching
lm_weight_ps <- lm_robust(trust_plebi_yes ~ treat_dicho, weight = IPW, data = estimation_nona) #0.04446

#controls matching
lm_weight_ps_controls <- lm_robust(trust_plebi_yes ~ treat_dicho + uribe_2002_proportion + uribe_2006_p
```

```
weight_model_ps <- weightit(balance_formula,
                      data = estimation_nona,
                      method = "glm",
                      estimand = "ATE")
```

```
## Warning: Some extreme weights were generated. Examine them with `summary()` and
## maybe trim them with `trim()`.
```

```
summary(weight_model_ps)
```

```
##                    Summary of weights
##
## - Weight ranges:
##
##             Min                                  Max
## treated 1.0000 ||                            19.6645
## control 1.0114 |---------------------------| 304.0845
##
## - Units with the 5 most extreme weights by group:
##
##               228     221     254     239      213
##  treated 11.3181 11.5045 13.1805 17.7038  19.6645
```

```
##                 78     832      33      81      830
##  control 12.5909 15.1573 18.2194 60.6592 304.0845
##
## - Weight statistics:
##
##         Coef of Var   MAD Entropy # Zeros
## treated       0.961 0.449   0.228       0
## control       5.002 0.846   1.239       0
##
## - Effective Sample Sizes:
##
##            Control Treated
## Unweighted  365.     641.
## Weighted     14.06   333.5
```

```
#treatment effect
summary(lm_weightit(trust_plebi_yes ~ treat_dicho, data = estimation_nona, weightit = weight_model_ps))
```

```
##
## Call:
## lm_weightit(formula = trust_plebi_yes ~ treat_dicho, data = estimation_nona,
##     weightit = weight_model_ps)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  0.42425    0.03996  10.618   <1e-06 ***
## treat_dicho  0.05344    0.04051   1.319    0.187
## Standard error: HC0 robust (adjusted for estimation of weights)
```

todo: check how extreme the weights get / how few units are doing most of the work - it means large bias

5.2 Kernel balancing weights

```
#Kbal at new defaults: 24 dimensions of k; -0.01064355
weight_model_kbal = kbal(allx = estimation_nona[, covariates],
                         treatment = estimation_nona$treat_dicho,
                         printprogress = TRUE)
```

```
## No multicollinear columns detected. Matrix is already full rank.
```

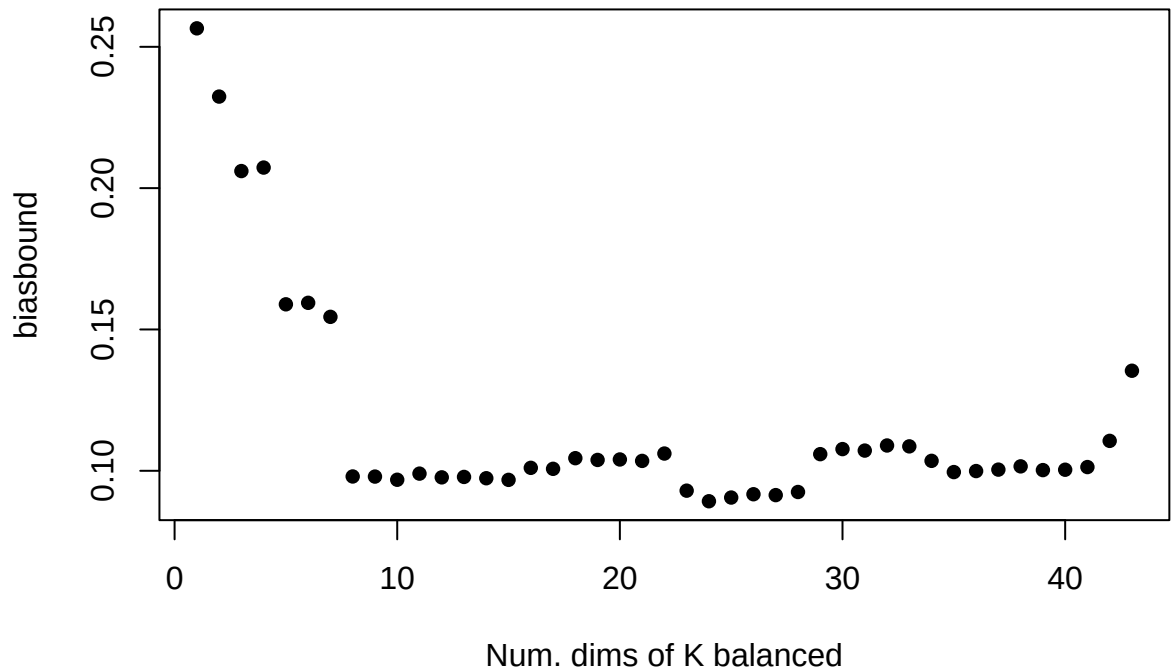
```
## Warning in kbal(allx = estimation_nona[, covariates], treatment =
## estimation_nona$treat_dicho, : One or more columns of "allx" contain less than
## 10 unique values, but "cat_data" and "mixed_data" are set to FALSE. Are you
## sure "allx" contains only continuous data?
```

```
## Searching for b value which maximizes the variance in K: 21.42 selected
## Building kernel matrix
## Running truncated SVD on kernel matrix up to 805 dimensions
## Truncated SVD with 805 first singular values accounts for 0.999965 of the variance of "K"
## Without balancing, biasbound (norm=1) is 0.24996 and the L1 discrepancy is 0.116
## With 1 dimensions of K, ebalance convergence is TRUE yielding biasbound (norm=1) of 0.22726
## With 2 dimensions of K, ebalance convergence is TRUE yielding biasbound (norm=1) of 0.20587
## With 3 dimensions of K, ebalance convergence is TRUE yielding biasbound (norm=1) of 0.18253
## With 4 dimensions of K, ebalance convergence is TRUE yielding biasbound (norm=1) of 0.18362
## With 5 dimensions of K, ebalance convergence is TRUE yielding biasbound (norm=1) of 0.14077
## With 6 dimensions of K, ebalance convergence is TRUE yielding biasbound (norm=1) of 0.14123
```

```
## With 7 dimensions of K, ebalance convergence is TRUE yielding biasbound (norm=1) of 0.13683
## With 8 dimensions of K, ebalance convergence is TRUE yielding biasbound (norm=1) of 0.08681
## With 9 dimensions of K, ebalance convergence is TRUE yielding biasbound (norm=1) of 0.08677
## With 10 dimensions of K, ebalance convergence is TRUE yielding biasbound (norm=1) of 0.08577
## With 11 dimensions of K, ebalance convergence is TRUE yielding biasbound (norm=1) of 0.0877
## With 12 dimensions of K, ebalance convergence is TRUE yielding biasbound (norm=1) of 0.08653
## With 13 dimensions of K, ebalance convergence is TRUE yielding biasbound (norm=1) of 0.08666
## With 14 dimensions of K, ebalance convergence is TRUE yielding biasbound (norm=1) of 0.08625
## With 15 dimensions of K, ebalance convergence is TRUE yielding biasbound (norm=1) of 0.08574
## With 16 dimensions of K, ebalance convergence is TRUE yielding biasbound (norm=1) of 0.08951
## With 17 dimensions of K, ebalance convergence is TRUE yielding biasbound (norm=1) of 0.0892
## With 18 dimensions of K, ebalance convergence is TRUE yielding biasbound (norm=1) of 0.09253
## With 19 dimensions of K, ebalance convergence is TRUE yielding biasbound (norm=1) of 0.09199
## With 20 dimensions of K, ebalance convergence is TRUE yielding biasbound (norm=1) of 0.09214
## With 21 dimensions of K, ebalance convergence is TRUE yielding biasbound (norm=1) of 0.09169
## With 22 dimensions of K, ebalance convergence is TRUE yielding biasbound (norm=1) of 0.09399
## With 23 dimensions of K, ebalance convergence is TRUE yielding biasbound (norm=1) of 0.08236
## With 24 dimensions of K, ebalance convergence is TRUE yielding biasbound (norm=1) of 0.07904
## With 25 dimensions of K, ebalance convergence is TRUE yielding biasbound (norm=1) of 0.0802
## With 26 dimensions of K, ebalance convergence is TRUE yielding biasbound (norm=1) of 0.08124
## With 27 dimensions of K, ebalance convergence is TRUE yielding biasbound (norm=1) of 0.08096
## With 28 dimensions of K, ebalance convergence is TRUE yielding biasbound (norm=1) of 0.08194
## With 29 dimensions of K, ebalance convergence is TRUE yielding biasbound (norm=1) of 0.0938
## With 30 dimensions of K, ebalance convergence is TRUE yielding biasbound (norm=1) of 0.09538
## With 31 dimensions of K, ebalance convergence is TRUE yielding biasbound (norm=1) of 0.09492
## With 32 dimensions of K, ebalance convergence is FALSE yielding biasbound (norm=1) of 0.09651
## With 33 dimensions of K, ebalance convergence is FALSE yielding biasbound (norm=1) of 0.09624
## With 34 dimensions of K, ebalance convergence is FALSE yielding biasbound (norm=1) of 0.0917
## With 35 dimensions of K, ebalance convergence is FALSE yielding biasbound (norm=1) of 0.08818
## With 36 dimensions of K, ebalance convergence is FALSE yielding biasbound (norm=1) of 0.0885
## With 37 dimensions of K, ebalance convergence is FALSE yielding biasbound (norm=1) of 0.08896
## With 38 dimensions of K, ebalance convergence is FALSE yielding biasbound (norm=1) of 0.08997
## With 39 dimensions of K, ebalance convergence is FALSE yielding biasbound (norm=1) of 0.0888
## With 40 dimensions of K, ebalance convergence is FALSE yielding biasbound (norm=1) of 0.08892
## With 41 dimensions of K, ebalance convergence is FALSE yielding biasbound (norm=1) of 0.08976
## With 42 dimensions of K, ebalance convergence is FALSE yielding biasbound (norm=1) of 0.09796
## With 43 dimensions of K, ebalance convergence is FALSE yielding biasbound (norm=1) of 0.11994
## Disregarding ebalance convergence and re-running at optimal choice of numdims, 24
```

```
lm_weight_ker <- lm_robust(trust_plebi_yes ~ treat_dicho, weights = weight_model_kbal$w, data = estimati
lm_weight_ker_controls <- lm_robust(trust_plebi_yes ~ treat_dicho + uribe_2002_proportion + uribe_2006_p
```

```
# Examine bias bound due to remaining imbalances;
# gamma is 55
# https://cran.r-project.org/web/packages/KRLS/KRLS.pdf

plot(x = weight_model_kbal$dist_record[1,],
     y = sd(estimation_nona$trust_plebi_yes) * sqrt(55) * weight_model_kbal$dist_record[2,],
     ylab="biasbound",
     xlab="Num. dims of K balanced",
     pch=16)
```



above ones compare means; kernel compares means, and also x/2, and x squared, etc etc; each one of these is a dimension of K; kbal finds the dimensions of k that makes the treatment and control the most balanced (ebalance convergence); we want bias bound to be as low as possible. The plot shows the number of K and what bias bound it gives; in this case, 24 is the lowest: "Disregarding ebalance convergence and re-running at optimal choice of numdims, 24"

```
rm(balance_table_ps, match_ps, match_balance_ps, match_balance_ps_2, match_data_ps, match_data_ps_2, we
```

6. Results

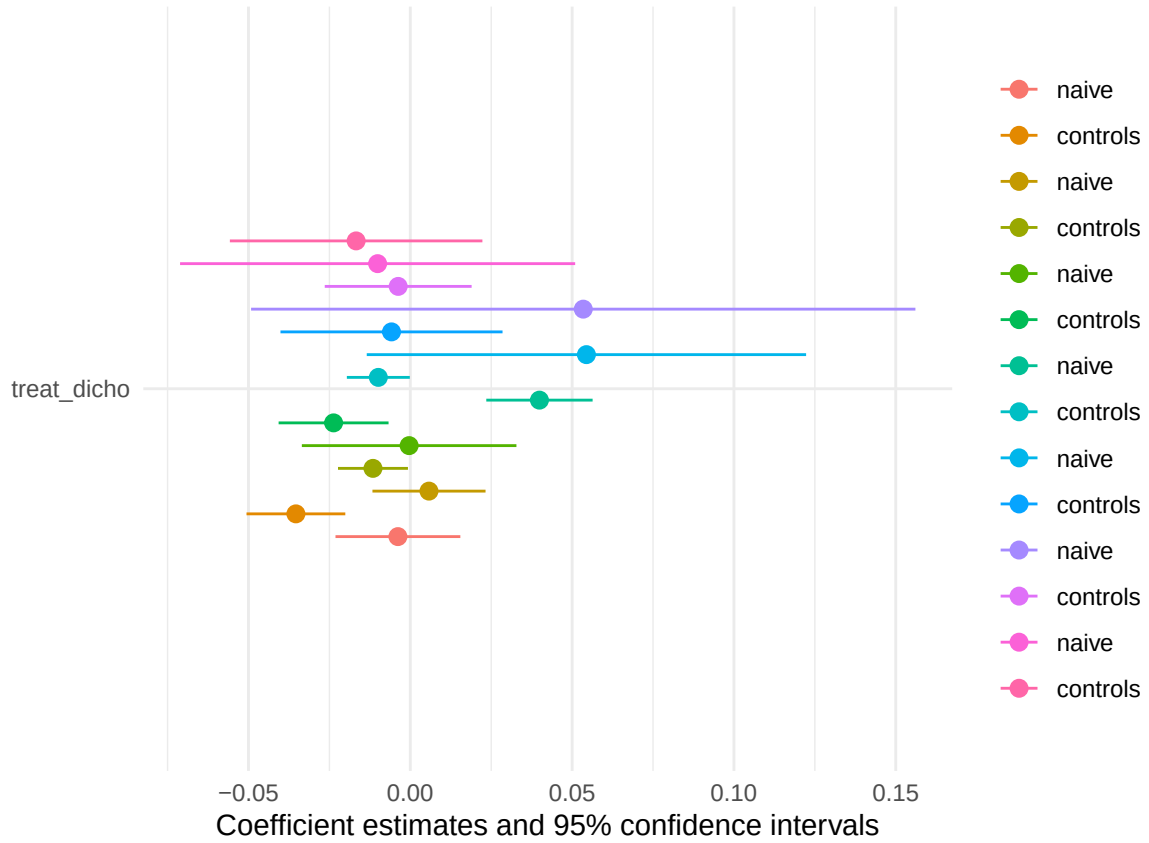
```
models <- list(
  "naive" = lm_naive,
  "controls" = lm_controls,
  "naive" = lm_match,
  "controls" = lm_match_controls,
  "naive" = lm_match2,
  "controls" = lm_match_controls2,
  "naive" = lm_match_ps,
  "controls" = lm_match_ps_controls,
  "naive" = lm_match_ps2,
  "controls" = lm_match_ps_controls2,
  "naive" = lm_weight_ps,
  "controls" = lm_weight_ps_controls,
  "naive" = lm_weight_ker,
  "controls" = lm_weight_ker_controls
  )
```

	Matching										Weights			
	Matching covariates						Matching propesity scores				Propensity Score		Kernel	
	Match()		MatchIt()				Match()		MatchIt()					
	naive	controls	naive	controls	naive	controls	naive	controls	naive	controls	naive	controls	naive	controls
(Intercept)	0.496***	0.561***	0.484***	0.593***	0.492***	0.522***	0.448***	0.735***	0.438***	0.629***	0.424***	0.722***	0.502***	0.364*
	(0.008)	(0.089)	(0.006)	(0.079)	(0.016)	(0.095)	(0.006)	(0.075)	(0.034)	(0.136)	(0.052)	(0.125)	(0.030)	(0.177)
treat_dichotomous	−0.004	−0.035***	0.006	−0.012*	0.000	−0.024**	0.040***	−0.010*	0.054	−0.006	0.053	−0.004	−0.010	−0.017
	(0.010)	(0.008)	(0.009)	(0.006)	(0.017)	(0.009)	(0.008)	(0.005)	(0.035)	(0.017)	(0.052)	(0.012)	(0.031)	(0.020)
uribe_2002_propensity		−0.099***		−0.141***		−0.090**		−0.150***		−0.136***		−0.117**		−0.052
		(0.026)		(0.026)		(0.029)		(0.022)		(0.037)		(0.040)		(0.066)
uribe_2006_propensity		−0.387***		−0.366***		−0.412***		−0.403***		−0.414***		−0.417***		−0.437***
		(0.024)		(0.025)		(0.026)		(0.022)		(0.027)		(0.030)		(0.035)
avg_schooling		0.034***		0.036***		0.038***		0.030***		0.033**		0.035***		0.035**
		(0.006)		(0.006)		(0.007)		(0.005)		(0.011)		(0.010)		(0.012)
literacy		0.002***		0.003***		0.003***		0.001***		0.002		0.001		0.003*
		(0.001)		(0.000)		(0.001)		(0.000)		(0.001)		(0.001)		(0.002)
log_pop		−0.006		−0.006		−0.004		−0.010**		−0.006		−0.015**		0.000
		(0.004)		(0.004)		(0.004)		(0.003)		(0.005)		(0.006)		(0.006)
prop_rural		0.031		0.019		0.039		0.013		0.027		0.031		0.030
		(0.023)		(0.022)		(0.026)		(0.019)		(0.035)		(0.031)		(0.043)
gdp_pcap_cons2005		−0.001		−0.001		−0.001		0.000		−0.001		0.000		−0.002+
		(0.001)		(0.001)		(0.001)		(0.001)		(0.002)		(0.001)		(0.001)
mpi		0.003***		0.003***		0.003***		0.003***		0.003***		0.003***		0.003***
		(0.000)		(0.000)		(0.000)		(0.000)		(0.000)		(0.000)		(0.001)
gini		−0.619***		−0.767***		−0.706***		−0.549***		−0.552***		−0.607***		−0.475*
		(0.108)		(0.101)		(0.118)		(0.092)		(0.142)		(0.151)		(0.217)
etv_state_pre		−0.001		−0.002+		−0.001		0.000		0.000		−0.002		0.000
		(0.001)		(0.001)		(0.001)		(0.001)		(0.002)		(0.002)		(0.002)
etv_nonstate_pre		0.000		0.000		0.000+		0.000***		0.000		0.000		0.000
		(0.000)		(0.000)		(0.000)		(0.000)		(0.000)		(0.000)		(0.000)
dem_security		0.022*		0.023*		0.023+		0.057***		0.048**		0.030		0.042*
		(0.011)		(0.011)		(0.012)		(0.010)		(0.016)		(0.019)		(0.016)
hist_land_conflict		−0.006		0.013		−0.008		0.003		−0.008		0.007		0.007
		(0.014)		(0.011)		(0.014)		(0.013)		(0.015)		(0.014)		(0.020)
Num.Obs.	1006	1006	1194	1194	849	849	1462	1462	799	799	1006	1006	1006	1006
R2	0.000	0.601	0.000	0.659	0.000	0.637	0.015	0.691	0.019	0.658	0.027	0.668	0.001	0.637
R2 Adj.	−0.001	0.595	0.000	0.655	−0.001	0.631	0.015	0.688	0.017	0.652	0.026	0.663	0.000	0.632
RMSE	0.15	0.10									0.16	0.10	0.15	0.10

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

```
modelsummary(models,
             #vcov = "HC2",
             #statistic = "p.value",
             stars = TRUE,
             gof_omit = "BIC|AIC|Log.Lik",
             output = "kableExtra") |>
  kable_styling(full_width = TRUE, position = "center", font_size = 8)|>
  add_header_above(c(" " = 3, "Match()" = 2, "MatchIt()" = 2, "Match()" = 2, "MatchIt()" = 2, " " = 4)
  add_header_above(c(" " = 3, "Matching covariates" = 4, "Matching propesity scores" = 4, "Propensity S
  add_header_above(c(" " = 3, "Matching" = 8, "Weights" = 4)) #|>
```

```
  #save_kable("code/outputs/etv_all_jep.png")

modelplot(models, coef_omit = c(-2))
```




```
#etv_jep, etv_all_jep, etv_cnmh, etv_all_cnmh, etv_false_pos
```

JEP: Adding controls makes it negative Propensity scores and kernel does not work well

cnmh: Adding controls makes it negative all matching and propensity score weighting looks good!

7. Sensitivity analysis

Hopefully using uribe 2002 as benchmark

```
#The previous estimate requires the assumption of no unobserved confounders for unbiasedness.
estimation_model <- lm(trust_plebi_yes ~ treat_dicho + uribe_2002_proportion + uribe_2006_proportion + a

estimation_sensitivity <- sensemakr(model = estimation_model,
                                    treatment = "treat_dicho",
                                    benchmark_covariates = "uribe_2002_proportion",
                                    kd = c(1, 5, 10, 15, 20), #1-3 times stronger than treatment
                                    ky = c(1, 5, 10, 15, 20), #1-3 times stronger than outcome
                                    q = 1, #reduction of 100% of current estimate - true effect of zero is p
                                    alpha = 0.05,
                                    reduce = T)
estimation_sensitivity
```

```
## Sensitivity Analysis to Unobserved Confounding
##
## Model Formula: trust_plebi_yes ~ treat_dicho + uribe_2002_proportion + uribe_2006_proportion +
##     avg_schooling + literacy + log_pop + prop_rural + gdp_pcap_cons2005 +
```

```
##     mpi + gini + etv_state_pre + etv_nonstate_pre + dem_security +
##     hist_land_conflict
##
## Null hypothesis: q = 1 and reduce = TRUE
##
## Unadjusted Estimates of ' treat_dicho ':
##   Coef. estimate: -0.03535
##   Standard Error: 0.00757
##   t-value: -4.66684
##
## Sensitivity Statistics:
##   Partial R2 of treatment with outcome: 0.0215
##   Robustness Value, q = 1 : 0.13767
##   Robustness Value, q = 1 alpha = 0.05 : 0.08227
##
## For more information, check summary.
```

```r
ovb_minimal_reporting(estimation_sensitivity, format = "latex")
```

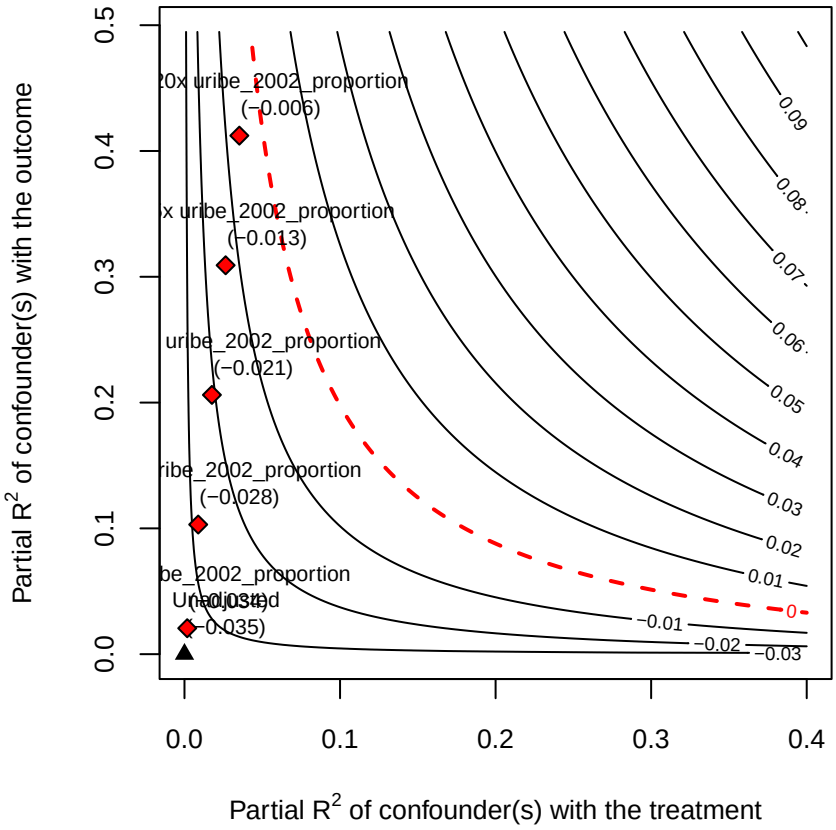
```
## \begin{table}[!h]
## \centering
## \begin{tabular}{lrrrrrr}
## \multicolumn{7}{c}{Outcome: \textit{trust_plebi_yes}} \\
## \hline \hline
## Treatment: & Est. & S.E. & t-value & $R^2_{Y \sim D |{\bf X}}$ & $RV_{q = 1}$ & $RV_{q = 1, \alpha =
## \hline
## \textit{treat_dicho} & -0.035 & 0.008 & -4.667 & 2.2\% & 13.8\% & 8.2\% \\
## \hline
## df = 991 & & \multicolumn{5}{r}{ \small \textit{Bound (1x uribe_2002_proportion)}: $R^2_{Y\sim Z| {\
## \end{tabular}
## \end{table}
```

```r
summary(estimation_sensitivity)
```

```
## Sensitivity Analysis to Unobserved Confounding
##
## Model Formula: trust_plebi_yes ~ treat_dicho + uribe_2002_proportion + uribe_2006_proportion +
##     avg_schooling + literacy + log_pop + prop_rural + gdp_pcap_cons2005 +
##     mpi + gini + etv_state_pre + etv_nonstate_pre + dem_security +
##     hist_land_conflict
##
## Null hypothesis: q = 1 and reduce = TRUE
## -- This means we are considering biases that reduce the absolute value of the current estimate.
## -- The null hypothesis deemed problematic is H0:tau = 0
##
## Unadjusted Estimates of 'treat_dicho':
##   Coef. estimate: -0.0353
##   Standard Error: 0.0076
##   t-value (H0:tau = 0): -4.6668
##
## Sensitivity Statistics:
##   Partial R2 of treatment with outcome: 0.0215
##   Robustness Value, q = 1: 0.1377
##   Robustness Value, q = 1, alpha = 0.05: 0.0823
##
```

```
## Verbal interpretation of sensitivity statistics:
##
## -- Partial R2 of the treatment with the outcome: an extreme confounder (orthogonal to the covariates)
##
## -- Robustness Value, q = 1: unobserved confounders (orthogonal to the covariates) that explain more t
##
## -- Robustness Value, q = 1, alpha = 0.05: unobserved confounders (orthogonal to the covariates) that
##
## Bounds on omitted variable bias:
##
## --The table below shows the maximum strength of unobserved confounders with association with the trea
##
##                 Bound Label R2dz.x R2yz.dx   Treatment Adjusted Estimate
##   1x uribe_2002_proportion 0.0018  0.0206 treat_dicho           -0.0339
##   5x uribe_2002_proportion 0.0088  0.1030 treat_dicho           -0.0281
##  10x uribe_2002_proportion 0.0176  0.2061 treat_dicho           -0.0208
##  15x uribe_2002_proportion 0.0265  0.3092 treat_dicho           -0.0135
##  20x uribe_2002_proportion 0.0353  0.4122 treat_dicho           -0.0061
##  Adjusted Se Adjusted T Adjusted Lower CI Adjusted Upper CI
##       0.0075    -4.5175           -0.0486           -0.0192
##       0.0072    -3.9020           -0.0423           -0.0140
##       0.0068    -3.0598           -0.0342           -0.0075
##       0.0064    -2.1140           -0.0260           -0.0010
##       0.0059    -1.0274           -0.0177            0.0055
```

```
plot(estimation_sensitivity)
```



sensitivity with ols only; ipw/matching weights mess it up

```
rm(estimation_model, estimation_sensitivity)
```

8. Robustness checks:

Falsos Positivos, State + Paramilitaries, And any kind of violence Changing time